UML DESIGN DOCUMENT

Travel Bridge Platform

Backend Design: Use Case, Class, Sequence, Activity, State, Architecture & ER Diagrams

Backend design views derived from the Travel Bridge requirements specification (32 features). Every diagram is traceable to one or more features.

Document version	1.0 (Draft for review)
Date	24 September 2026
Based on	Travel Bridge Platform – Requirements Specification, v1.0
Diagrams	46 diagrams (source in PlantUML, PNG and SVG provided separately)
Audience	Backend developers, architects, QA, product owner

1. Introduction

This document collects the design-level UML views of the Travel Bridge backend. The diagrams turn the feature-based requirements into a shared technical picture that developers can implement, reviewers can challenge, and testers can use to derive scenarios.

1.1 Diagram Types Included

Diagram type	Answers the question	Count
Use Case	Who uses the system and for what?	6
Class	What are the domain concepts, their behavior, and how is the code layered and extended?	8
Sequence	How do components collaborate at runtime for the important flows?	13
Activity	What are the workflows and the journey-generation algorithm?	4
State Machine	Which states can an entity be in and how does it move between them?	8
Package / Component / Deployment	How is the system structured, connected, and deployed?	3
Entity-Relationship	How is data stored and constrained? (not UML, but required for the backend)	4

Object, communication, timing, interaction-overview, and composite-structure diagrams were considered and left out: they would repeat information already shown in the diagrams above.

1.2 Conventions

- Diagram IDs use a type prefix (UC, CLS, SEQ, ACT, ST, ARCH, ERD). Feature IDs (F-01 to F-32) refer to the requirements specification.
- Class diagrams use C#-style types (Guid, decimal, DateTime); ERD types use PostgreSQL names.
- Statuses are stored as small integers mapped to enums; timestamps are stored in UTC.
- «ref» classes are shown only to anchor relationships to classes defined in another diagram.
- Large diagrams are placed on A3 pages for legibility. Each diagram is also delivered as a separate PNG and SVG file.

2. Diagram Catalog

Diagram names link to their page. Feature references use the IDs from the requirements specification.

ID	Diagram	Traces to features
Part A – Use Case Diagrams		
UC-01	Visitor – Public Access	F-01, F-03, F-04, F-06–F-08, F-20–F-24
UC-02	Tourist – Profile, Community & Guides	F-02, F-09–F-11, F-18, F-19, F-29
UC-03	Tourist – AI & Trip Planning	F-12–F-17, F-27, F-28
UC-04	Tour Guide	F-18, F-19, F-25, F-29
UC-05	Administrator – Content Management	F-03–F-06, F-20, F-22–F-24
UC-06	Administrator – Moderation, People & Platform	F-10, F-11, F-18, F-29–F-32
Part B – Class Diagrams (Domain Model & Design)		
CLS-01	Domain Model – Identity, Geography & Catalog	F-01–F-06, F-26
CLS-02	Domain Model – Community, Moderation & Notifications	F-09–F-11, F-25, F-29
CLS-03	Domain Model – AI Assistant, Journey, Trip & Cost	F-12–F-17
CLS-04	Domain Model – Tour Guides, Requests & Chat	F-18, F-19
CLS-05	Domain Model – Transportation, Culture, Safety & Awareness	F-20, F-22–F-24
CLS-06	Layered Architecture – Destination Module Example	F-06–F-08
CLS-07	Extensibility Patterns (1/2) – AI Provider & Recommendations	F-12, F-13, F-16, F-17
CLS-08	Extensibility Patterns (2/2) – Notifications & Currency	F-25, F-27
Part C – Sequence Diagrams		
SEQ-01	Registration, Login & Token Refresh	F-01
SEQ-02	Search, Filter & Sort Destinations	F-07
SEQ-03	Review Submission & Admin Moderation	F-10, F-25
SEQ-04	Place Suggestion to Draft Destination	F-06, F-11
SEQ-05	AI Assistant – Requirement Gathering	F-12
SEQ-06	Journey Generation	F-13, F-14
SEQ-07	Journey Customization via Natural Language	F-13, F-14
SEQ-08	Trip Cost Estimation	F-14, F-27
SEQ-09	Guide Request, Decision & Chat	F-19, F-25
SEQ-10	AI Image Recognition & Price Estimate	F-17, F-27
SEQ-11	Notification Delivery (Transactional Outbox)	F-25
SEQ-12	Reporting & Moderation	F-10, F-29
SEQ-13	Guide Onboarding & Admin Approval	F-01, F-18
Part D – Activity Diagrams		
ACT-01	Core Tourist Journey (End to End)	F-01, F-02, F-07, F-09, F-10, F-12–F-15, F-19
ACT-02	Journey Generation Algorithm	F-13, F-14

ID	Diagram	Traces to features
ACT-03	Guide Request Workflow	F-18, F-19
ACT-04	Place Suggestion to Published Destination	F-06, F-11
Part E – State Machine Diagrams		
ST-01	Review	F-10, F-29
ST-02	Place Suggestion	F-11
ST-03	Destination	F-06
ST-04	Guide Profile	F-18
ST-05	Guide Request	F-19
ST-06	Trip (Lifecycle and Visibility)	F-15
ST-07	Report	F-29
ST-08	User Account	F-01, F-31
Part F – Architecture Diagrams		
ARCH-01	Package Diagram – Modular Monolith	F-01, F-03, F-06, F-10, F-13, F-18, F-25, F-30
ARCH-02	Component Diagram – Logical Architecture	F-01, F-12, F-19, F-25, F-27, F-30
ARCH-03	Deployment Diagram – Proposed Topology	F-30
Part G – Entity-Relationship Diagrams		
ERD-01	Identity, Geography & Catalog	F-01–F-06, F-26
ERD-02	Community, Moderation, Notifications & Audit	F-09–F-11, F-25, F-29
ERD-03	Conversations, Journeys, Trips, Cost & AI	F-12–F-17, F-27
ERD-04	Guides, Chat, Transportation, Culture & Safety	F-18–F-20, F-22–F-24

3. Key Design Decisions

These decisions shape the diagrams. They are proposals for review, not fixed constraints.

Decision	Rationale	Shown in
Modular monolith	One deployable unit keeps the MVP simple, while strict module boundaries allow later extraction into services.	ARCH-01, ARCH-02
Clean layering per module	API → Application → Domain, with Infrastructure implementing interfaces, keeps business rules testable and independent of frameworks.	CLS-06
JWT access tokens with rotating refresh tokens	Short-lived access tokens limit exposure; refresh-token rotation and reuse detection protect sessions. Roles are data-driven.	SEQ-01, ST-08
Transactional outbox	Events are stored with the business change, so notifications are never lost and are delivered at least once.	SEQ-11, CLS-08
AI behind an abstraction	An adapter with decorators isolates the vendor, adds rate limiting and usage tracking, and lets the domain validate everything the AI proposes.	CLS-07, SEQ-05 – SEQ-07
AI grounded in the catalog	Journeys use only Published destinations. The AI supplies intent and wording, never places or prices.	SEQ-06, ACT-02
Search on the relational store with caching	Indexes and spatial types are enough for the MVP; tagged cache invalidation keeps results fresh. A dedicated search engine can be added later.	SEQ-02, ERD-01
Translation tables with English fallback	New languages need no schema changes and missing text never breaks a page.	CLS-01, ERD-01
Prices stored in original currency	Conversion happens at read time using cached rates with last-known fallback, so history is never distorted.	SEQ-08, CLS-08
Soft delete, archive, and audit trail	Content referenced by reviews, favorites, and trips is never physically removed; admin actions are traceable.	CLS-01, ERD-02
Optimistic concurrency and partial unique index	Prevents two guides' sessions or requests from colliding without heavy locking.	SEQ-09, ERD-04
Polymorphic references for reports and rules	One report or rule table serves many targets; the cost is application-level integrity checks instead of foreign keys.	CLS-02, CLS-05
Background worker	Outbox delivery, request expiry, and exchange-rate refresh run outside web requests.	ARCH-02, ARCH-03

3.1 Suggested Technology Mapping

Concern	Suggested option
API host	ASP.NET Core Web API with OpenAPI documentation and SignalR for chat
Data access	Entity Framework Core (migrations, soft-delete query filters, optimistic concurrency)
Validation and mapping	FluentValidation and a mapper for DTOs
Database	PostgreSQL with PostGIS (or SQL Server with spatial types)
Cache and rate limits	Redis

Concern	Suggested option
Files	S3-compatible or Azure Blob storage behind IFileStorage
Background jobs	ASP.NET Core hosted services or Hangfire
Observability	Structured logging and OpenTelemetry traces and metrics
Testing	xUnit or NUnit for unit and integration tests

Part A – Use Case Diagrams

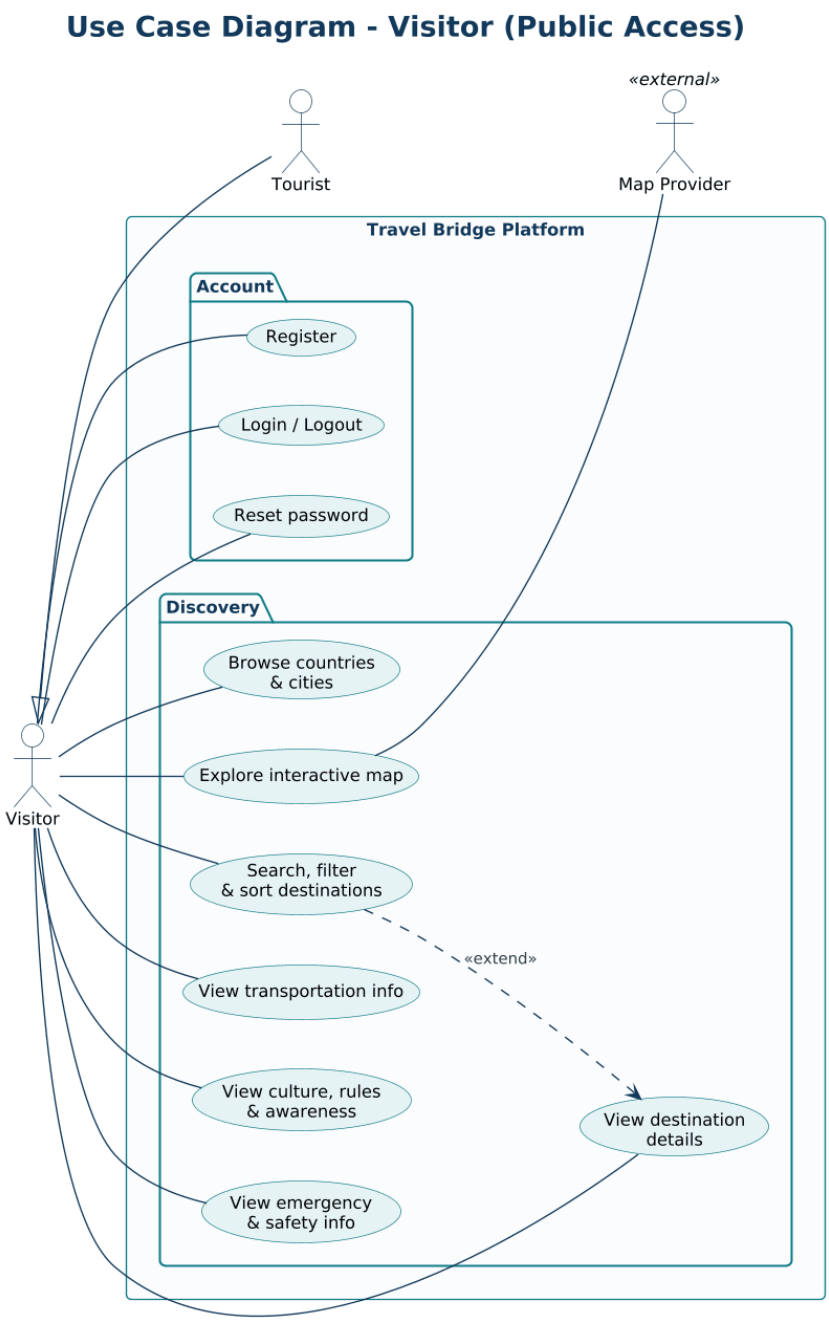
Actors and the capabilities they use, grouped so each diagram stays readable.

UC-01 Visitor – Public Access

Shows what an unauthenticated visitor can do: sign up or log in and browse public tourism content. A Tourist inherits every Visitor capability.

Traces to: F-01, F-03, F-04, F-06–F-08, F-20–F-24

- Discovery works without an account; only personal features require sign-in.
- Map tiles and geocoding come from an external map provider.

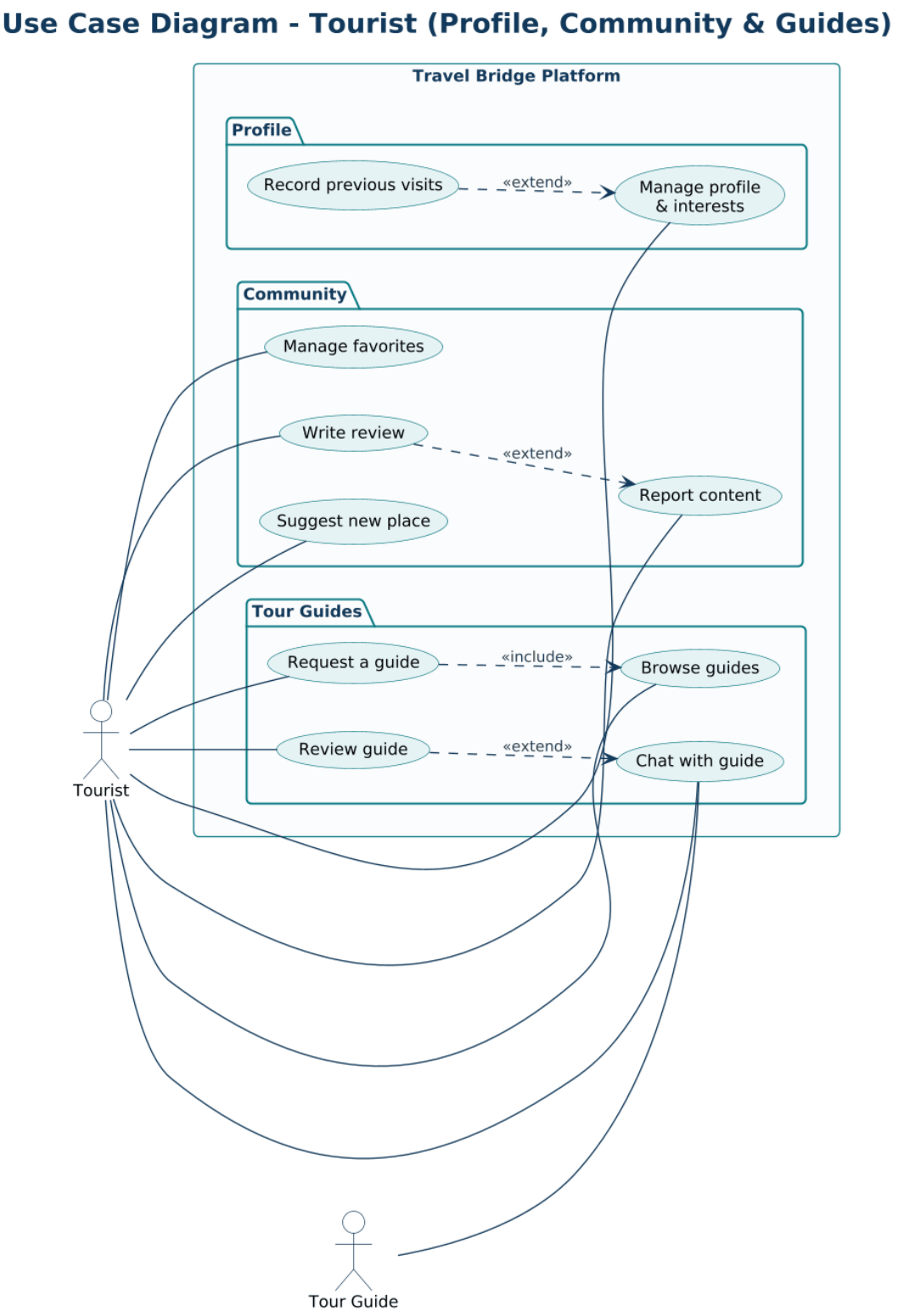


UC-02 Tourist – Profile, Community & Guides

Authenticated tourist capabilities for personalization, community contribution, and working with tour guides.

Traces to: F-02, F-09–F-11, F-18, F-19, F-29

- All use cases here require the Tourist role.
- Chat with a guide exists only after a guide request is accepted.

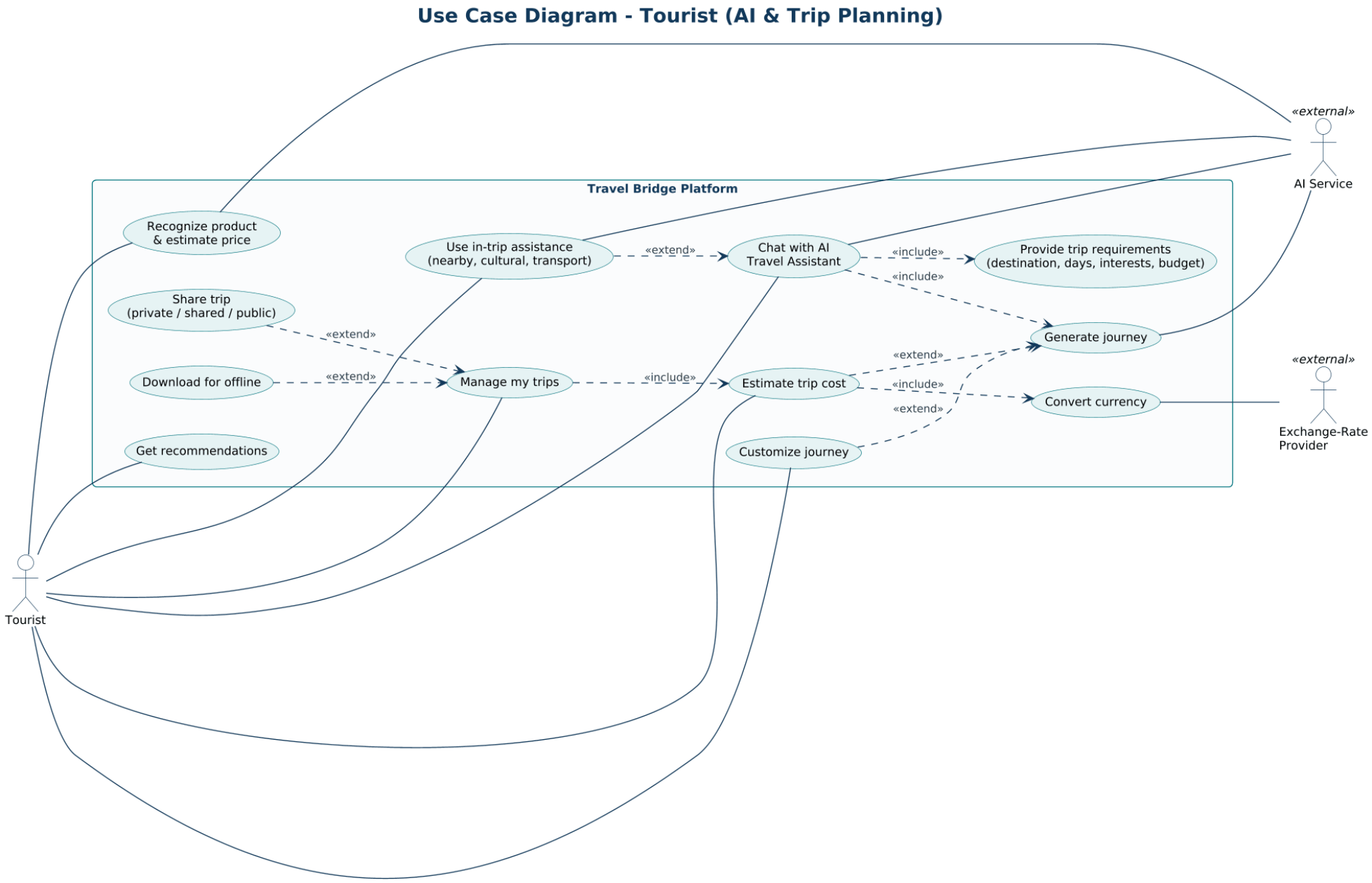


UC-03 Tourist – AI & Trip Planning

AI-assisted planning: conversation, journey generation and customization, cost estimation, trip management, sharing, offline use, and image recognition.

Traces to: F-12–F-17, F-27, F-28

- AI output is always validated by the platform before it is saved.
- Currency conversion uses a cached exchange-rate source.

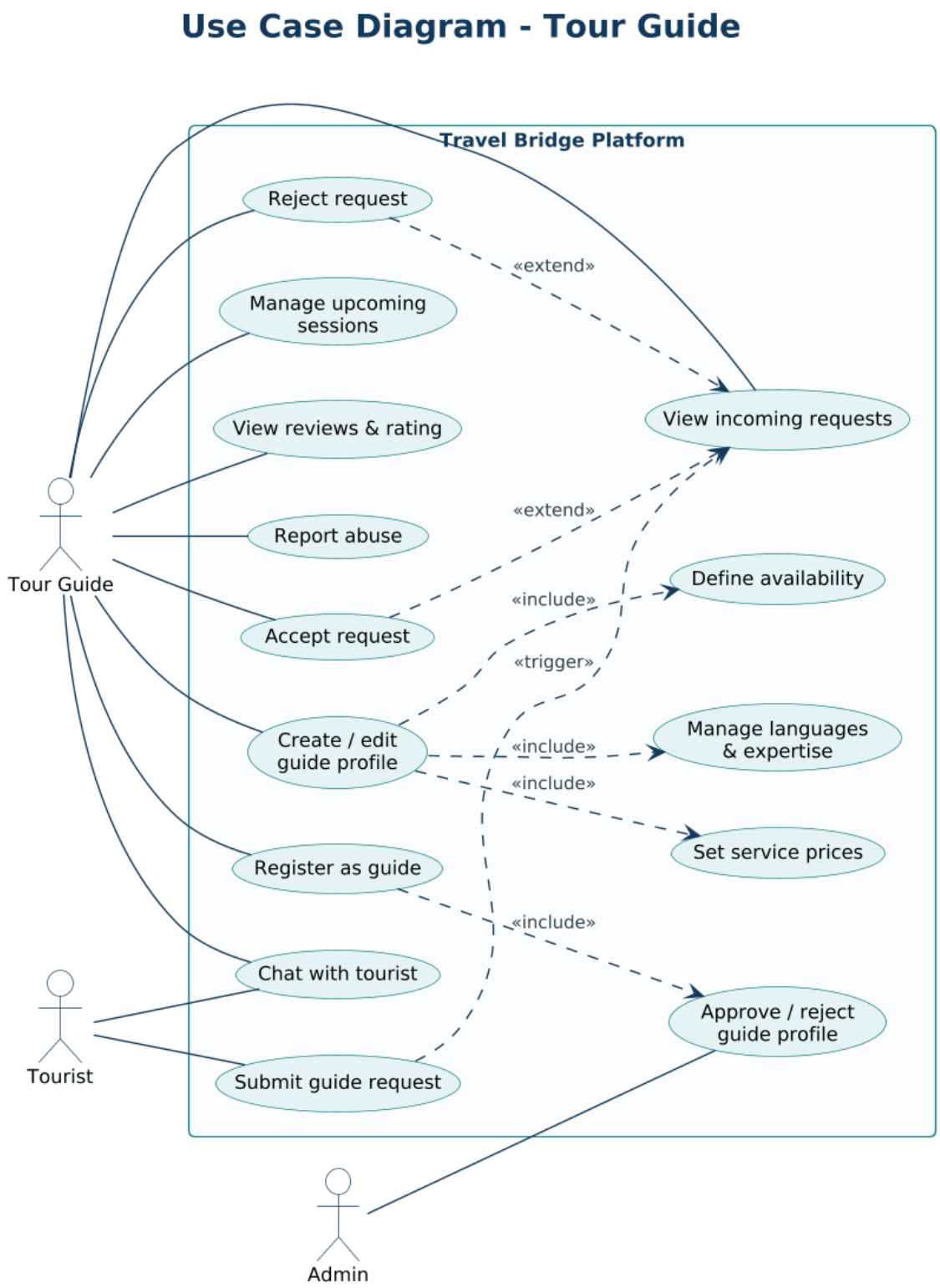


UC-04 Tour Guide

Guide capabilities: profile and availability management, handling incoming requests, chatting with tourists, and managing sessions.

Traces to: F-18, F-19, F-25, F-29

- A guide can act only after an admin approves the profile.
- Requests are created by tourists and consumed by guides.

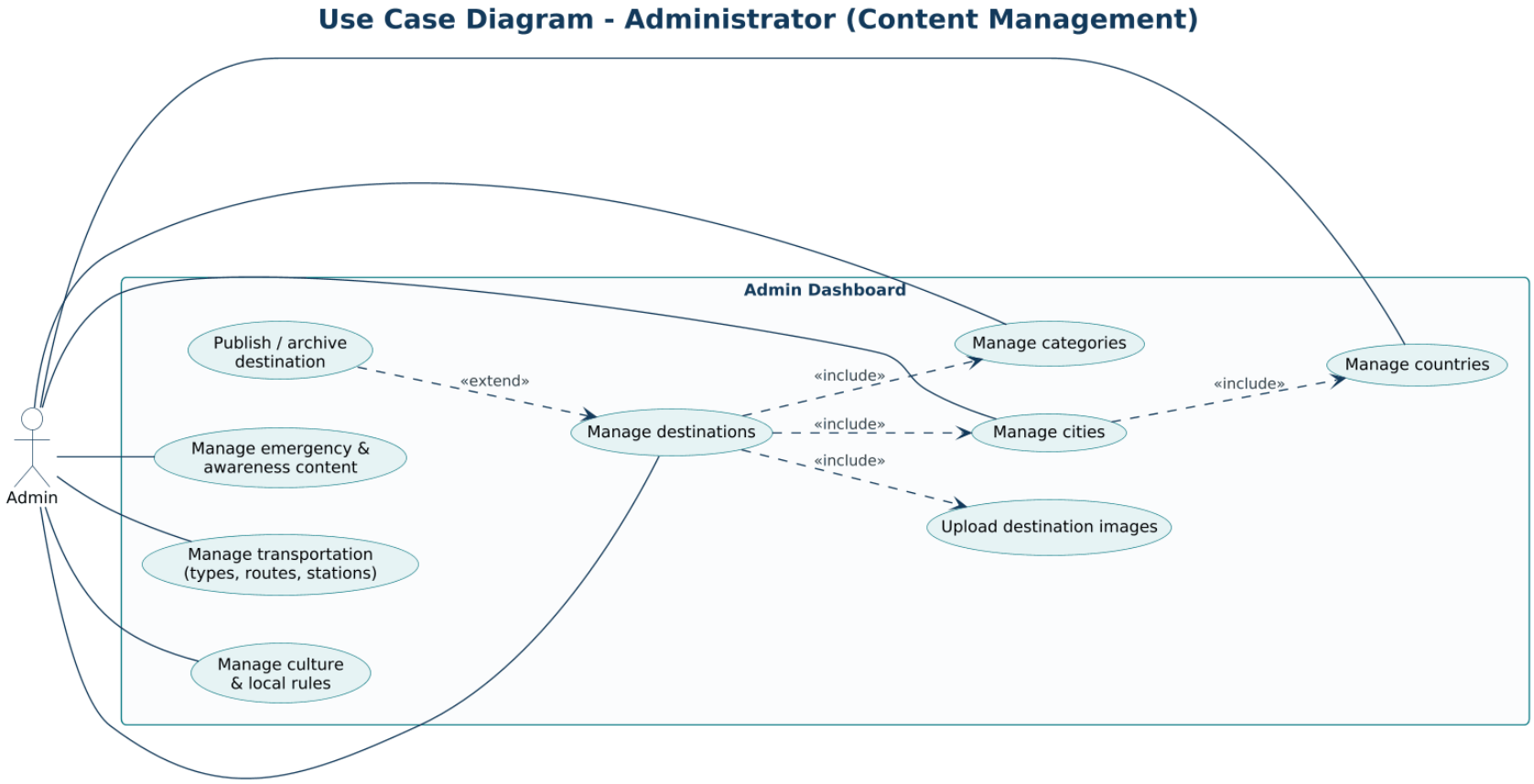


UC-05 Administrator – Content Management

Admin management of the tourism content hierarchy and supporting information.

Traces to: F-03–F-06, F-20, F-22–F-24

- A destination needs an existing city and category and at least one image before it can be published.
- Delete is implemented as archive.



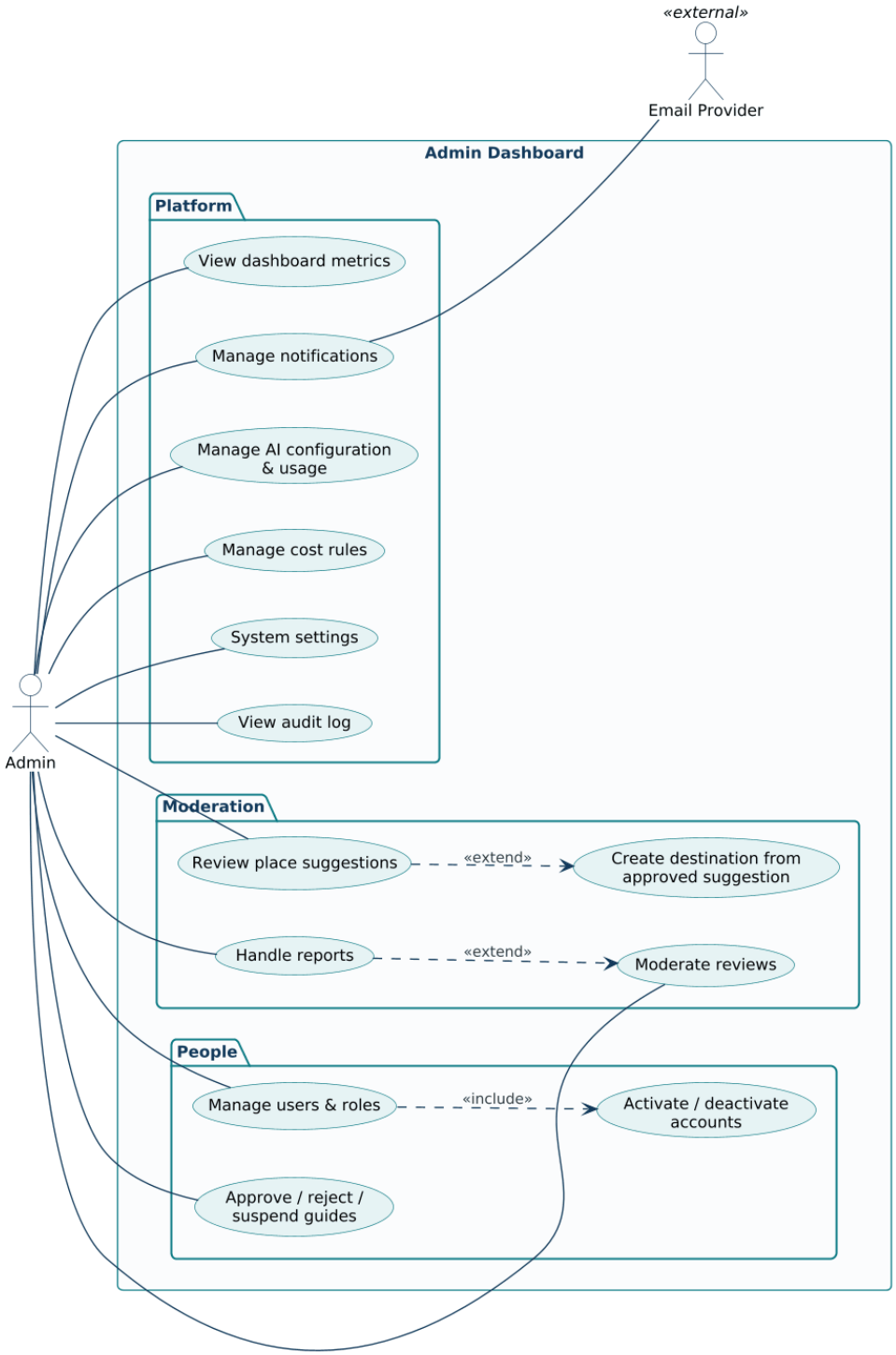
UC-06 Administrator – Moderation, People & Platform

Admin moderation, user and guide management, and platform-level configuration and monitoring.

Traces to: F-10, F-11, F-18, F-29–F-32

- Approving a place suggestion leads to creating a Draft destination.
- Every admin action is written to the audit log.

Use Case Diagram - Administrator (Moderation, People & Platform)



Part B – Class Diagrams (Domain Model & Design)

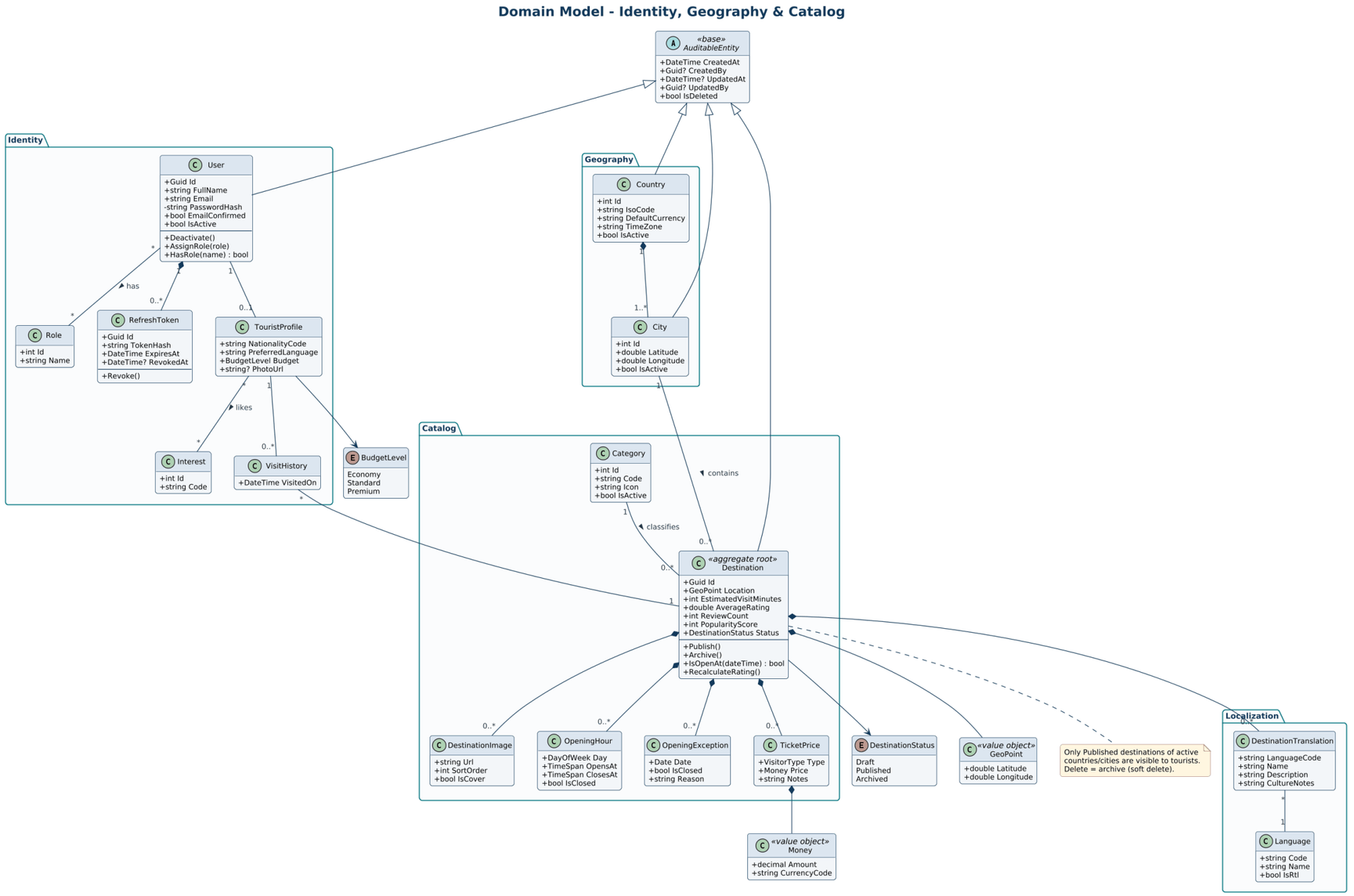
Domain classes with attributes, behavior, and relationships, plus the layered structure and extensibility patterns used by the backend.

CLS-01 Domain Model – Identity, Geography & Catalog

Core domain model for users, roles, tourist profiles, the Country → City → Destination hierarchy, categories, and localized destination content.

Traces to: F-01–F-06, F-26

- All aggregates inherit audit fields and soft delete from AuditableEntity.
- Translations live in separate tables so new languages need no schema change.

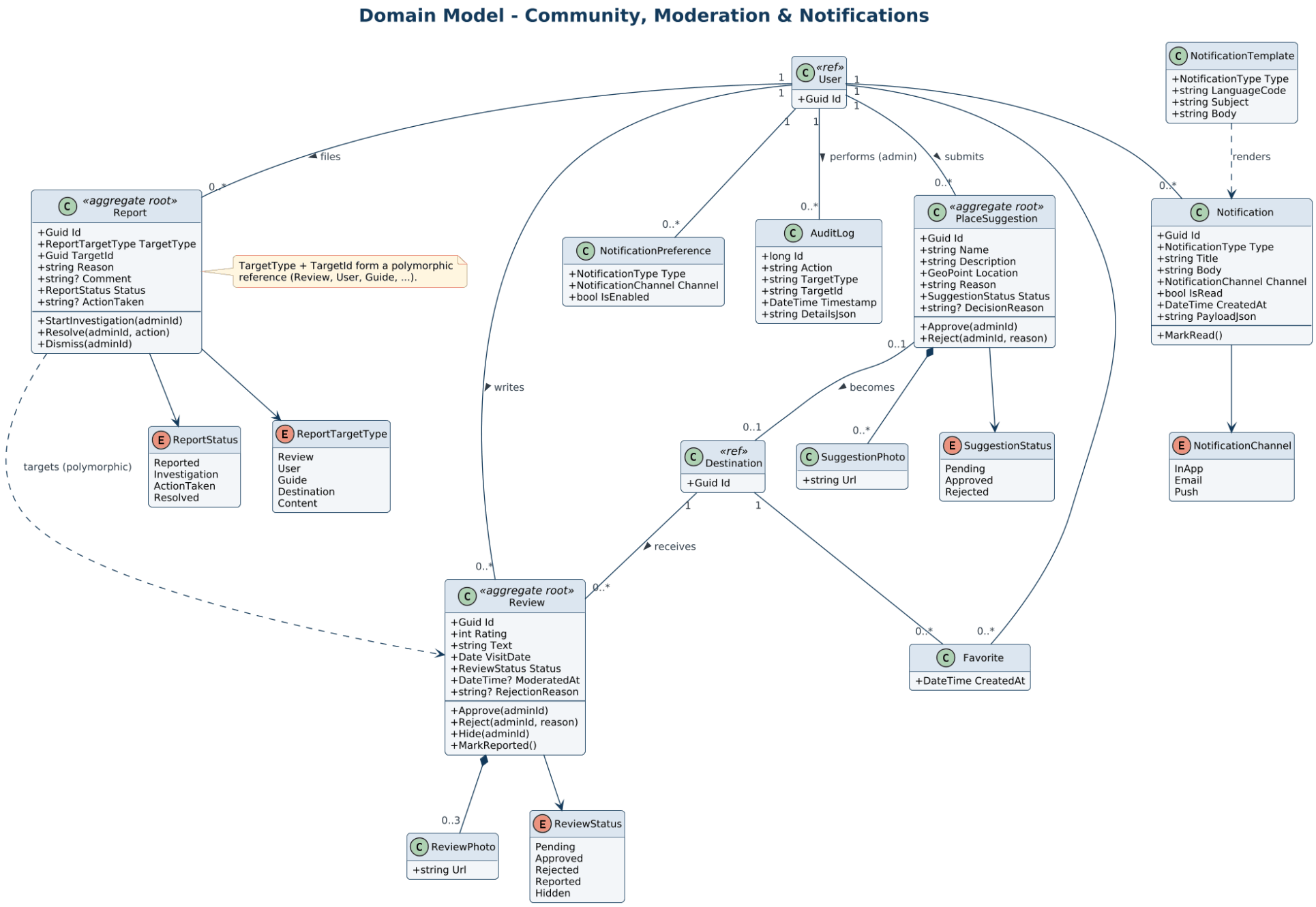


CLS-02 Domain Model – Community, Moderation & Notifications

Favorites, reviews, place suggestions, reports, notifications, and the audit log.

Traces to: F-09–F-11, F-25, F-29

- Report uses a polymorphic (TargetType, TargetId) reference to cover reviews, users, guides, and destinations.
- Review, PlaceSuggestion, and Report own their state transitions.

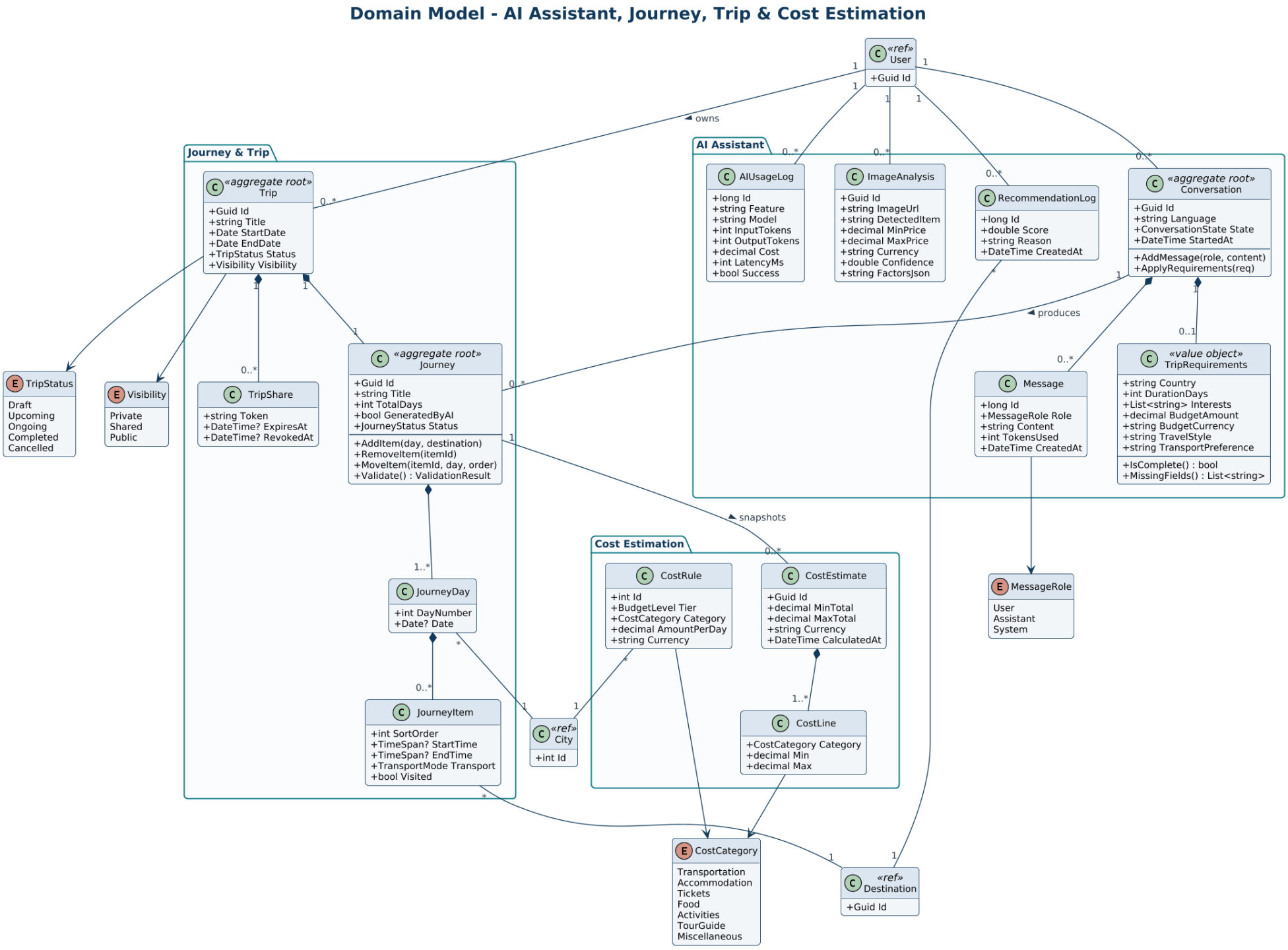


CLS-03 Domain Model – AI Assistant, Journey, Trip & Cost

Conversation and requirement extraction, journey structure, trip lifecycle and sharing, cost estimation, image analysis, and AI usage tracking.

Traces to: F-12–F-17

- Journey is the aggregate that guarantees itinerary validity; Trip adds dates, status, and visibility.
- CostEstimate rows are immutable, timestamped snapshots.

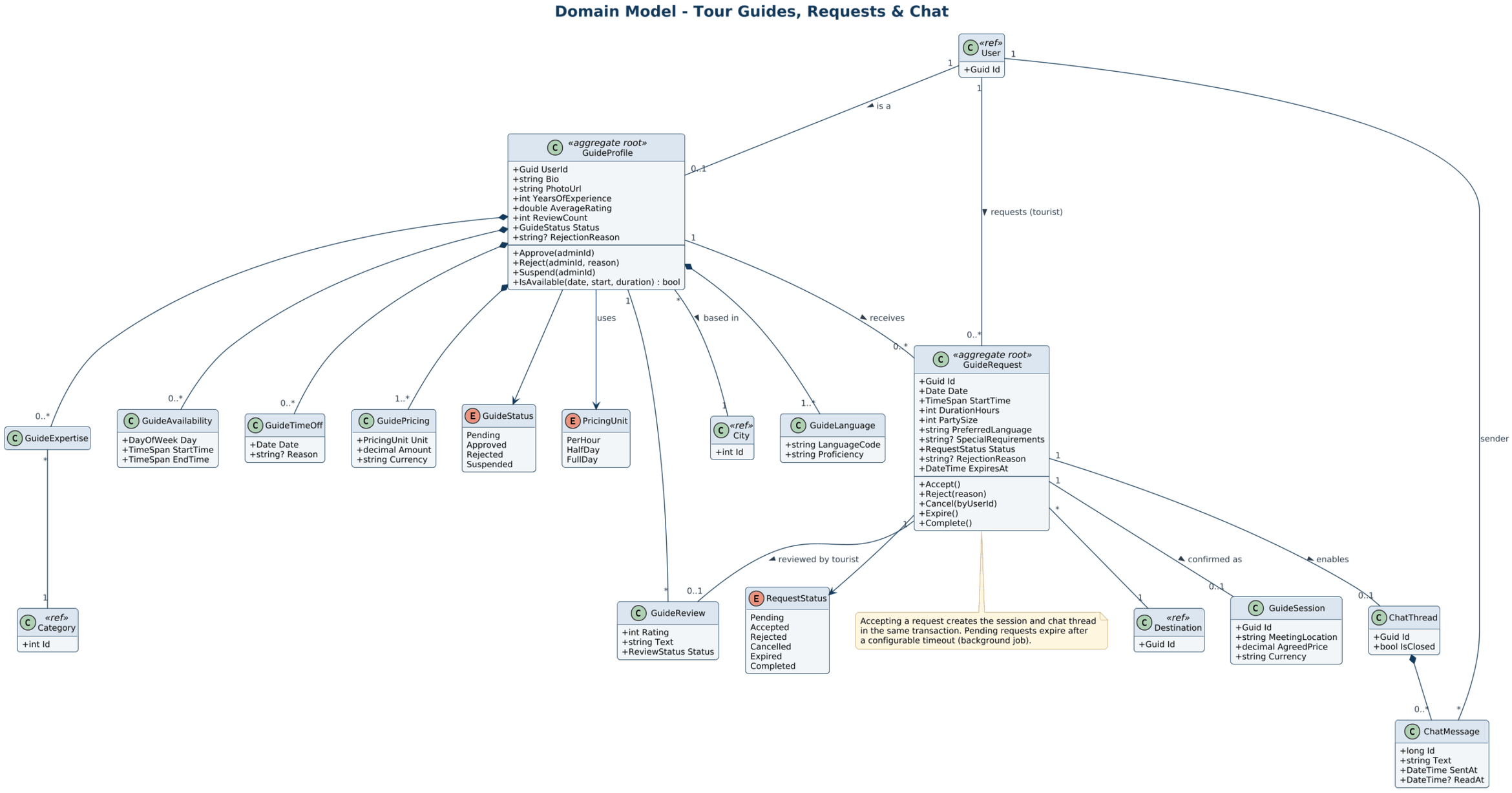


CLS-04 Domain Model – Tour Guides, Requests & Chat

Guide profiles with languages, expertise, availability, and pricing; guide requests; sessions; chat; and guide reviews.

Traces to: F-18, F-19

- GuideRequest owns accept, reject, cancel, and expire behavior.
- Accepting creates the session and chat thread in the same transaction.

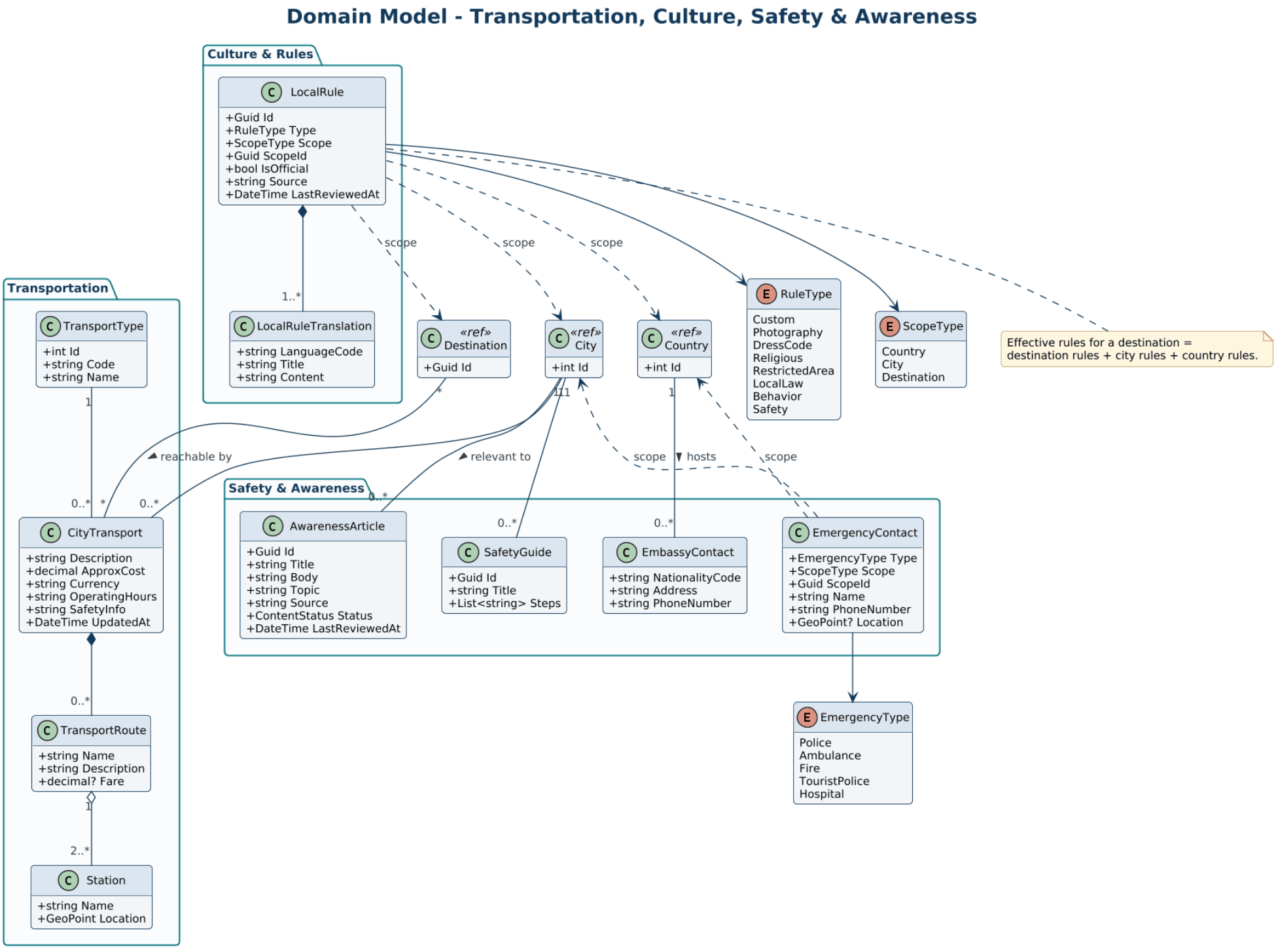


CLS-05 Domain Model – Transportation, Culture, Safety & Awareness

City transportation, local rules and customs, emergency contacts, embassies, safety guides, and awareness articles.

Traces to: F-20, F-22–F-24

- Rules and emergency contacts attach to a country, city, or destination through (Scope, ScopeId).
- Effective rules for a destination combine destination, city, and country rules.

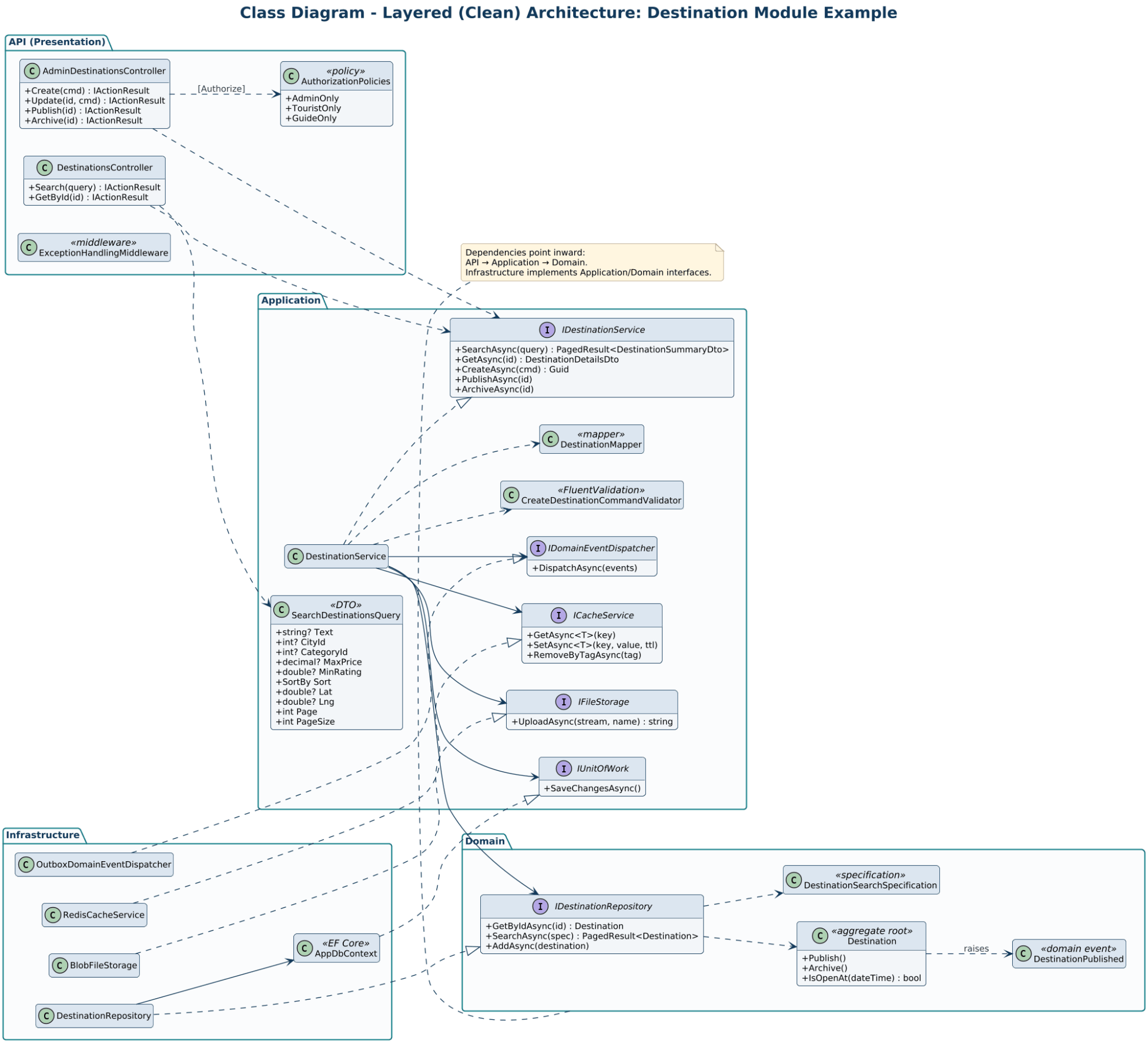


CLS-06 Layered Architecture – Destination Module Example

Shows how one module is layered: controllers, application services, domain aggregate and repository interfaces, and infrastructure implementations.

Traces to: F-06–F-08

- Dependencies point inward: API → Application → Domain; Infrastructure implements the interfaces.
- Every module follows the same structure.



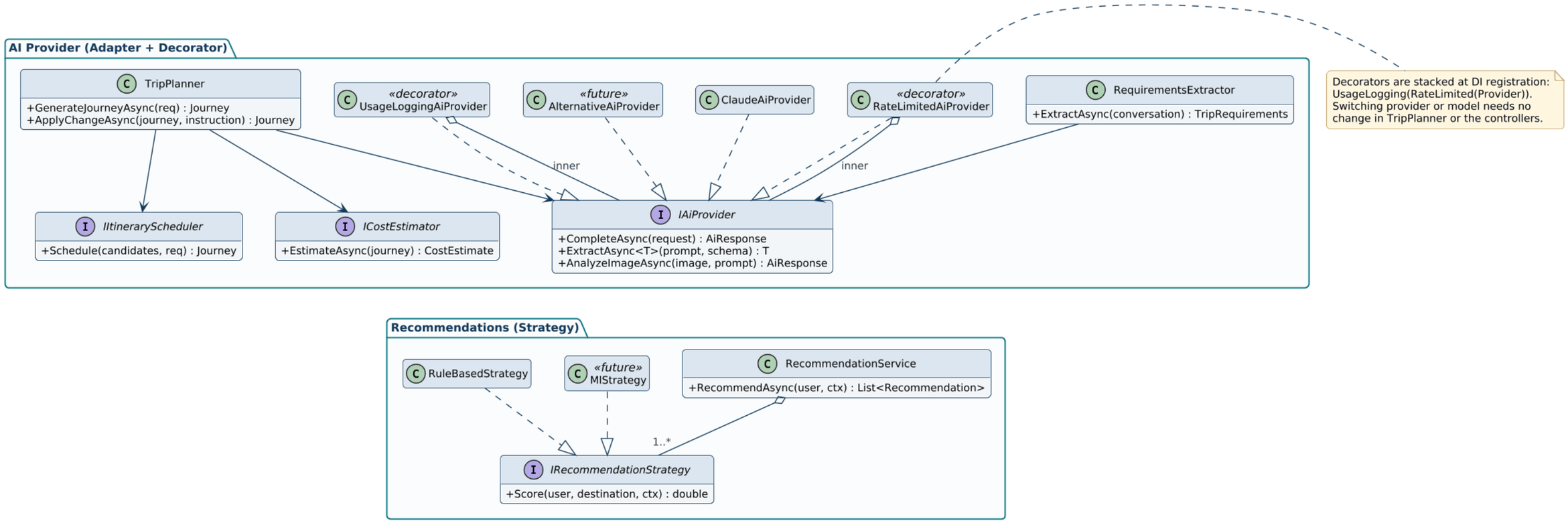
CLS-07 Extensibility Patterns (1/2) – AI Provider & Recommendations

Adapter and decorator around the AI provider, and a strategy pattern for recommendations.

Traces to: F-12, F-13, F-16, F-17

- Usage logging and rate limiting are decorators, so features never depend on a vendor.
- New recommendation strategies plug in without changing RecommendationService.

Class Diagram - Extensibility Patterns (1/2): AI Provider & Recommendations



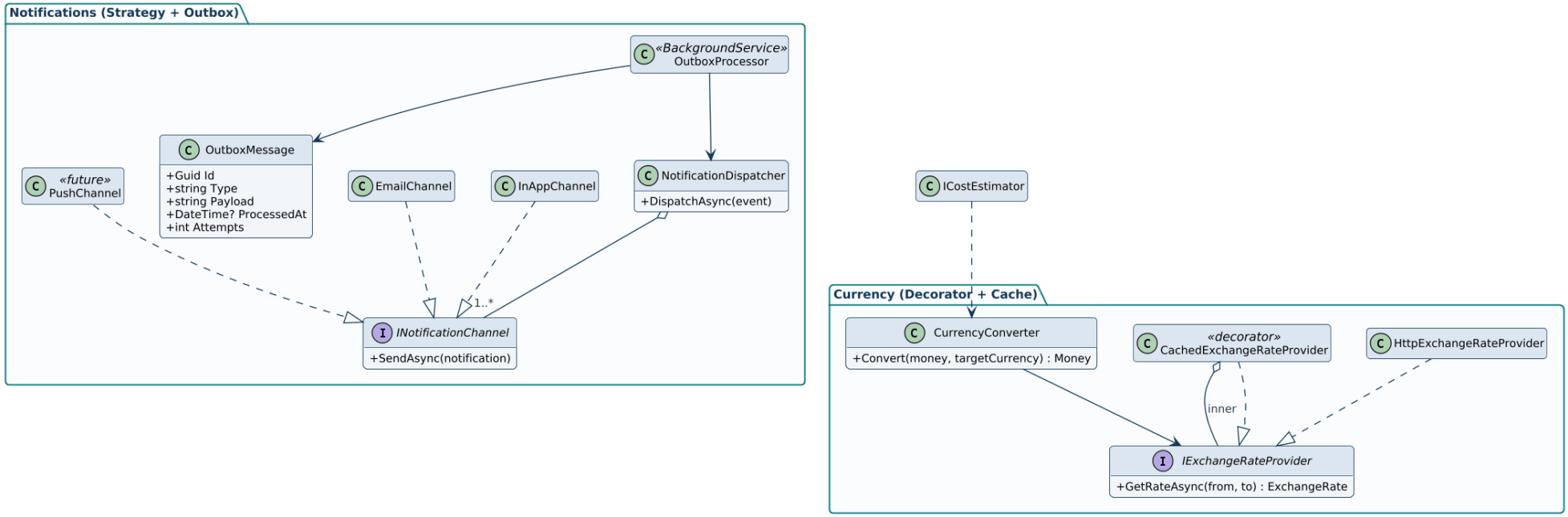
CLS-08 Extensibility Patterns (2/2) – Notifications & Currency

Pluggable notification channels with a transactional outbox, and a cached exchange-rate provider.

Traces to: F-25, F-27

- The outbox guarantees at-least-once delivery of events.
- CachedExchangeRateProvider falls back to the last known rate when the provider is down.

Class Diagram - Extensibility Patterns (2/2): Notifications & Currency



Part C – Sequence Diagrams

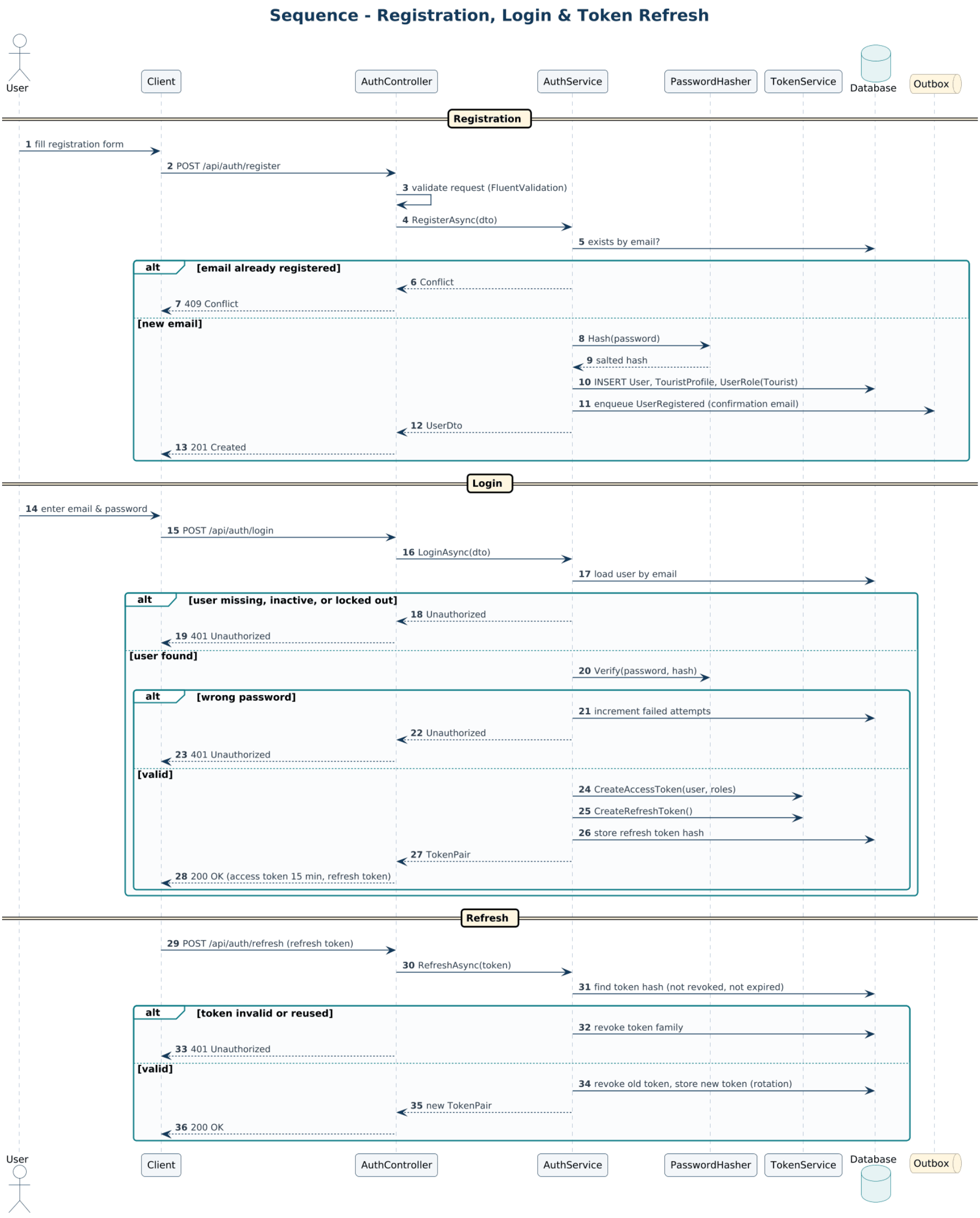
Key runtime interactions between clients, API, services, database, and external systems.

SEQ-01 Registration, Login & Token Refresh

Account creation with hashed passwords, JWT-based login, and refresh-token rotation.

Traces to: F-01

- Access tokens are short-lived; refresh tokens are stored hashed, rotated on use, and revoked when reuse is detected.
- Confirmation emails are sent asynchronously through the outbox.

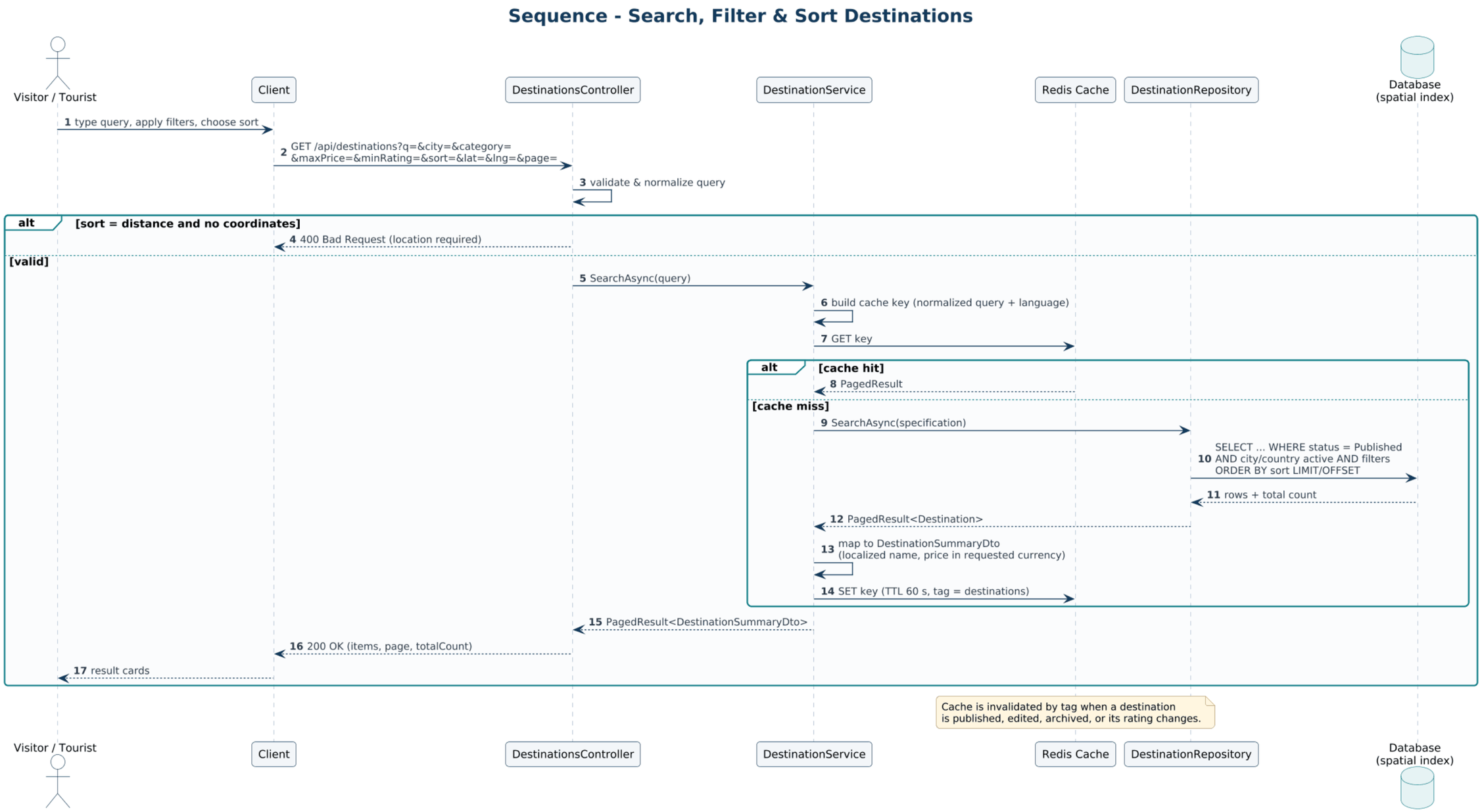


SEQ-02 Search, Filter & Sort Destinations

Request flow for destination search including cache lookup, filtered database query, and response mapping.

Traces to: F-07

- Only Published destinations in active cities and countries are searched.
- Cache entries are tagged and invalidated when content or ratings change.

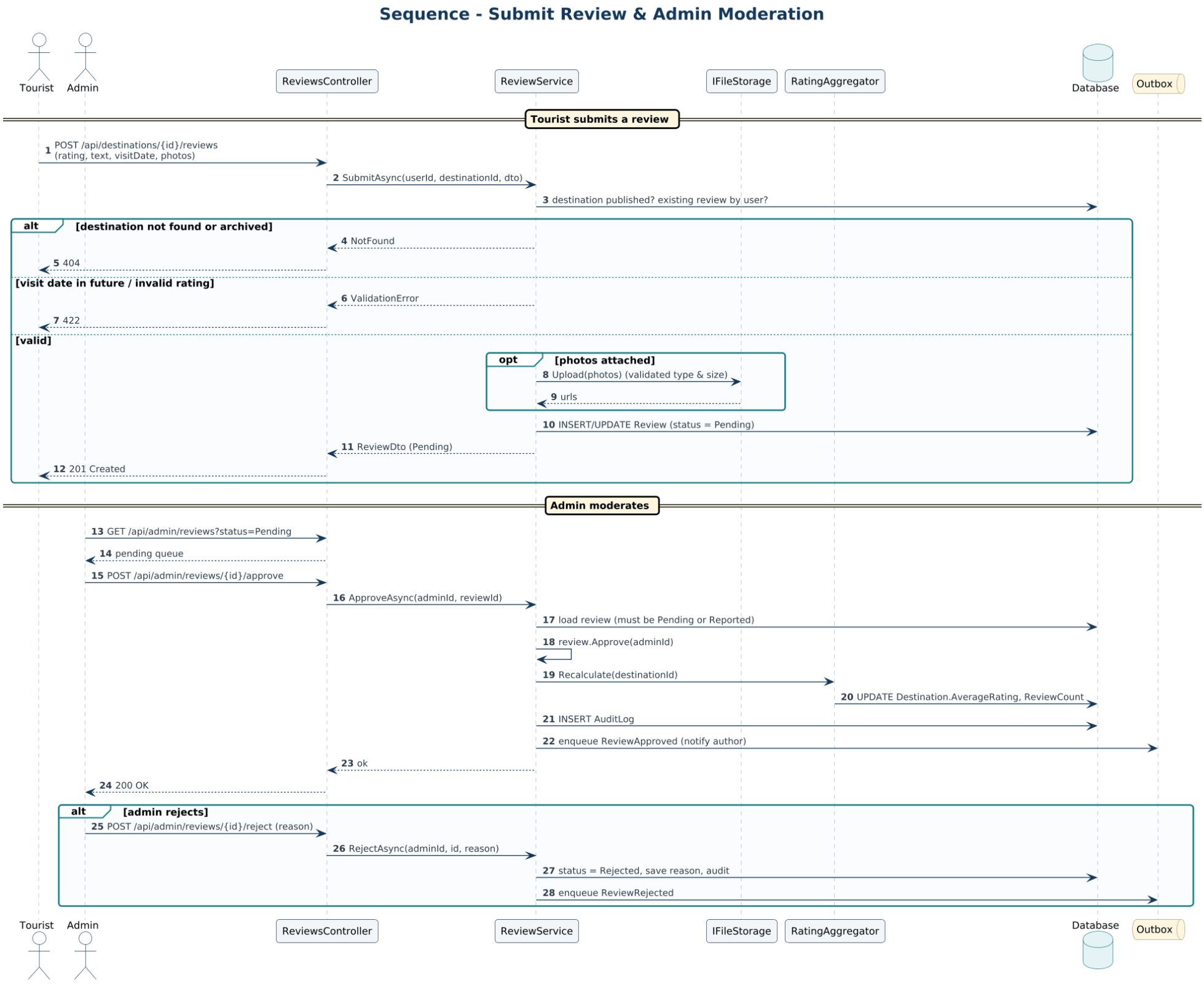


SEQ-03 Review Submission & Admin Moderation

A tourist submits a review, then an admin approves or rejects it, updating the destination rating.

Traces to: F-10, F-25

- Reviews start as Pending and become public only after approval.
- Rating aggregates are recalculated in the same request as the approval.

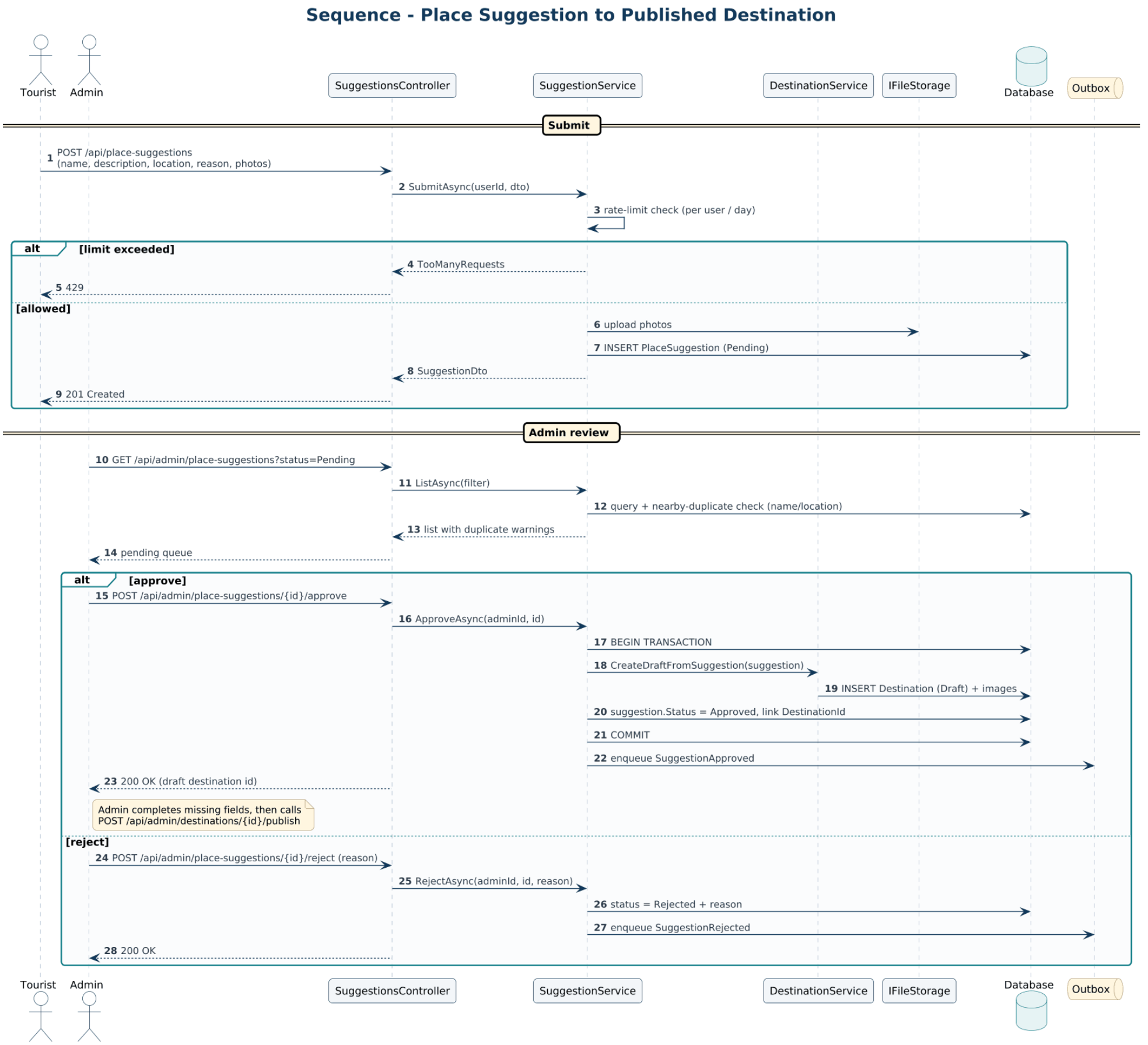


SEQ-04 Place Suggestion to Draft Destination

Submission, duplicate detection, admin decision, and creation of a Draft destination.

Traces to: F-06, F-11

- Approval and destination creation happen in one transaction.
- Publishing is a separate, explicit admin step.

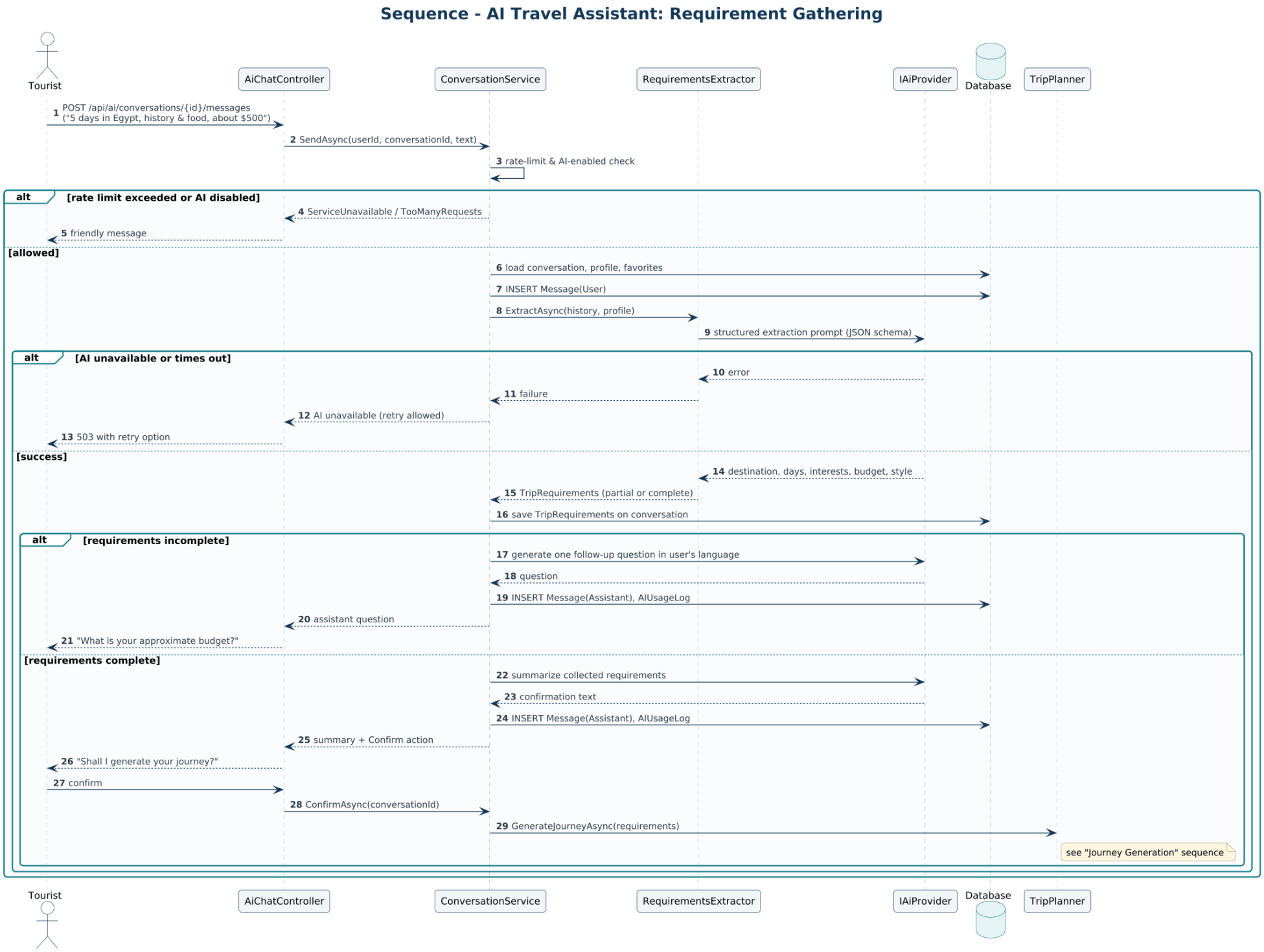


SEQ-05 AI Assistant – Requirement Gathering

How the assistant extracts trip requirements, asks for missing details, and hands over to journey generation.

Traces to: F-12

- One question is asked at a time; requirements are confirmed before generating.
- AI failure produces a retryable error instead of a broken conversation.

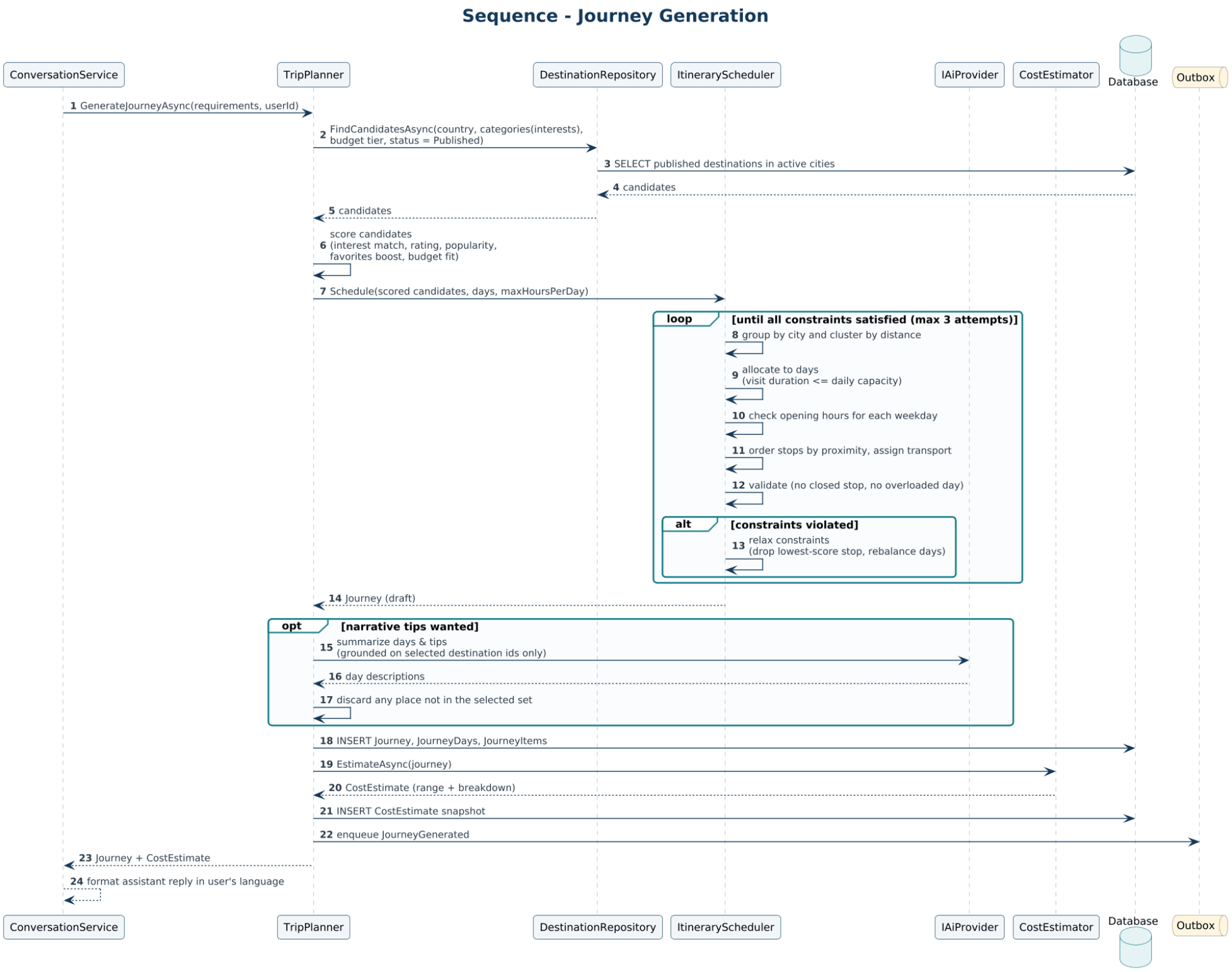


SEQ-06 Journey Generation

Candidate selection, scheduling under constraints, optional AI narrative, persistence, and cost estimate.

Traces to: F-13, F-14

- Places always come from the catalog; the AI only adds text about the selected destinations.
- The scheduler enforces opening hours and daily capacity.

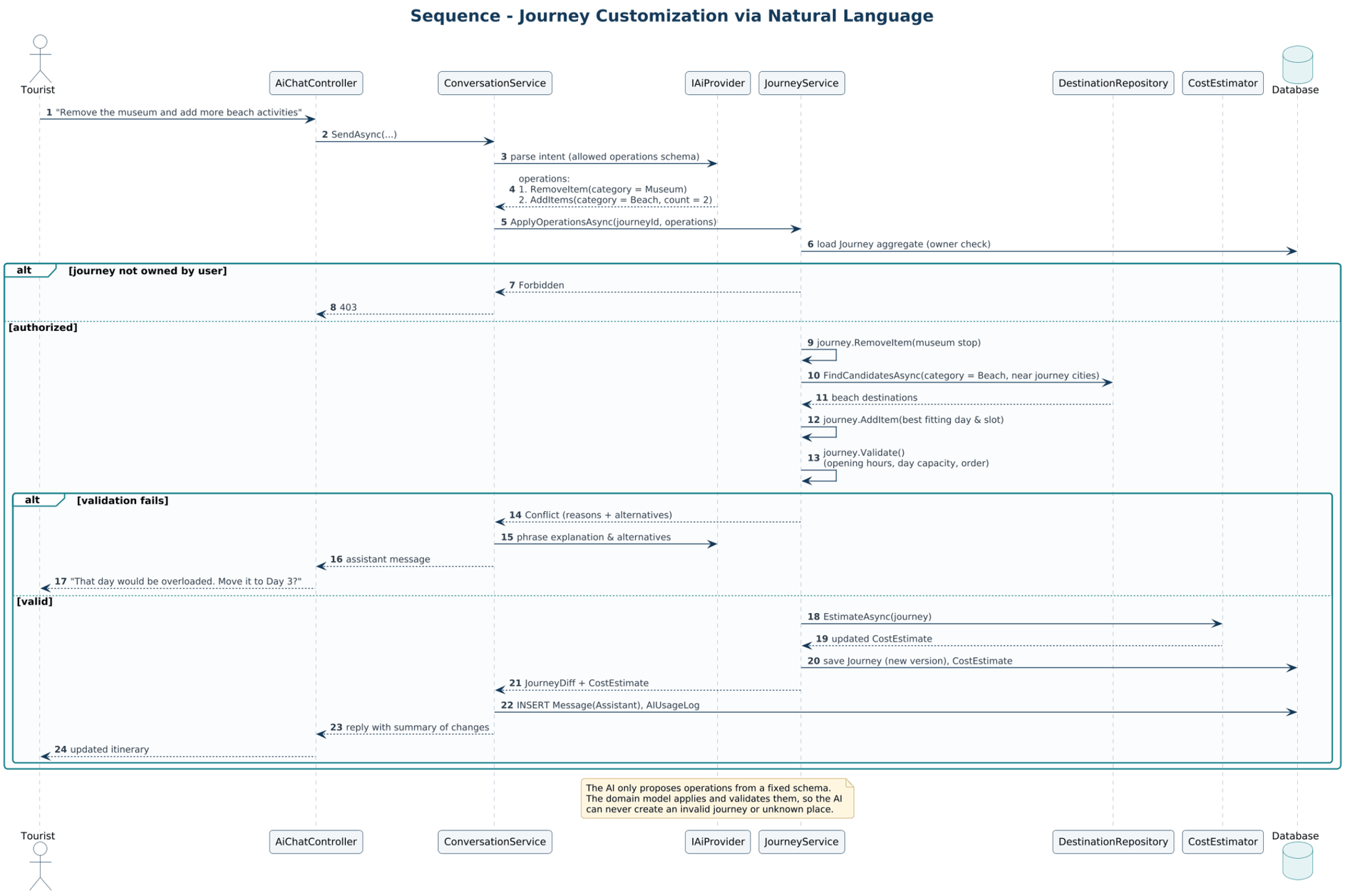


SEQ-07 Journey Customization via Natural Language

Turning a free-text change request into validated journey operations.

Traces to: F-13, F-14

- The AI proposes operations from a fixed schema; the domain model applies and validates them.
- Conflicts return alternatives instead of silently breaking the itinerary.

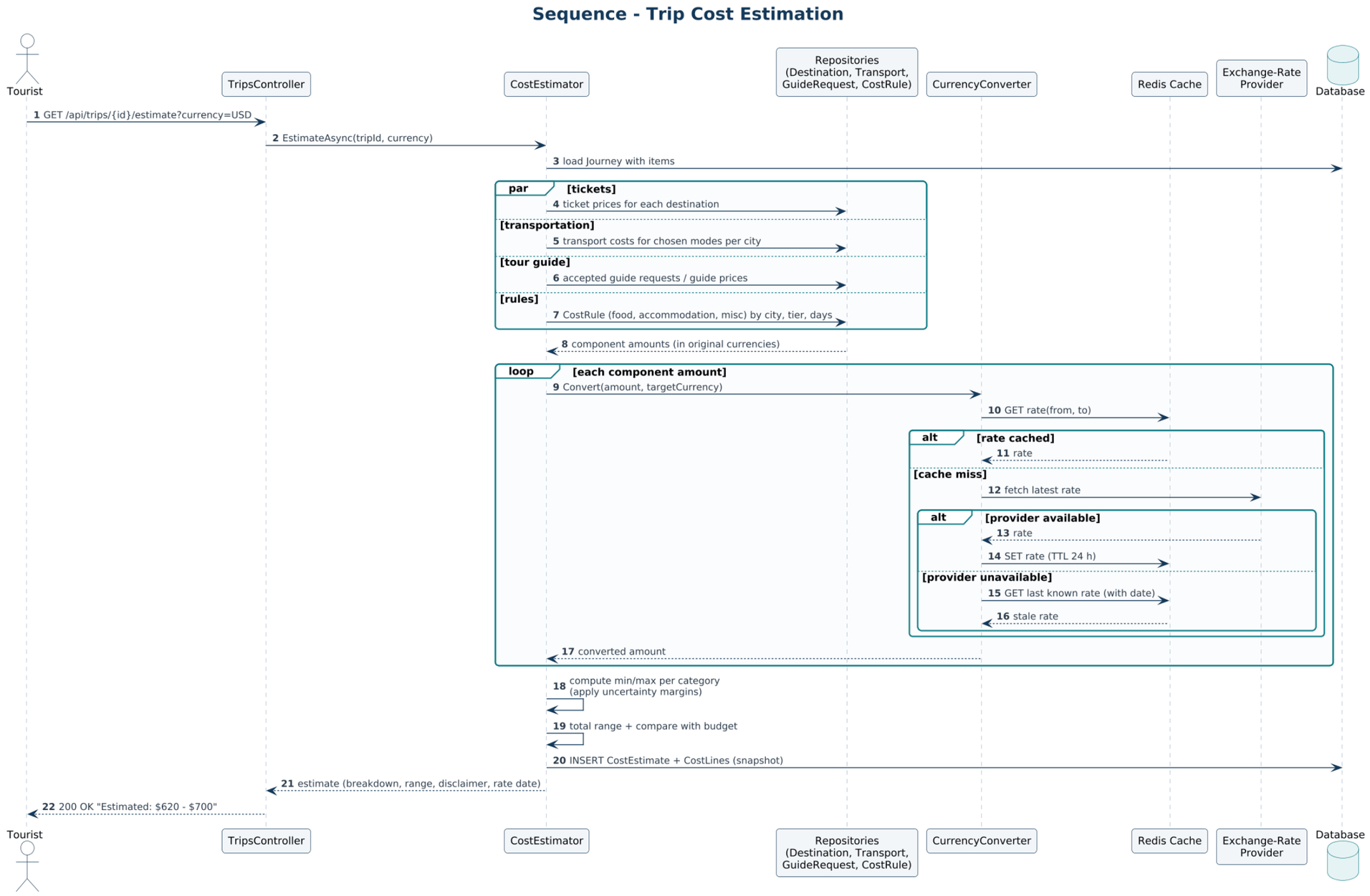


SEQ-08 Trip Cost Estimation

Computing an estimate range from tickets, transport, guide costs, and admin-configured rules, with currency conversion.

Traces to: F-14, F-27

- Cost components are gathered in parallel and returned as a range with a disclaimer.
- Exchange rates fall back to the last known value when the provider is unavailable.

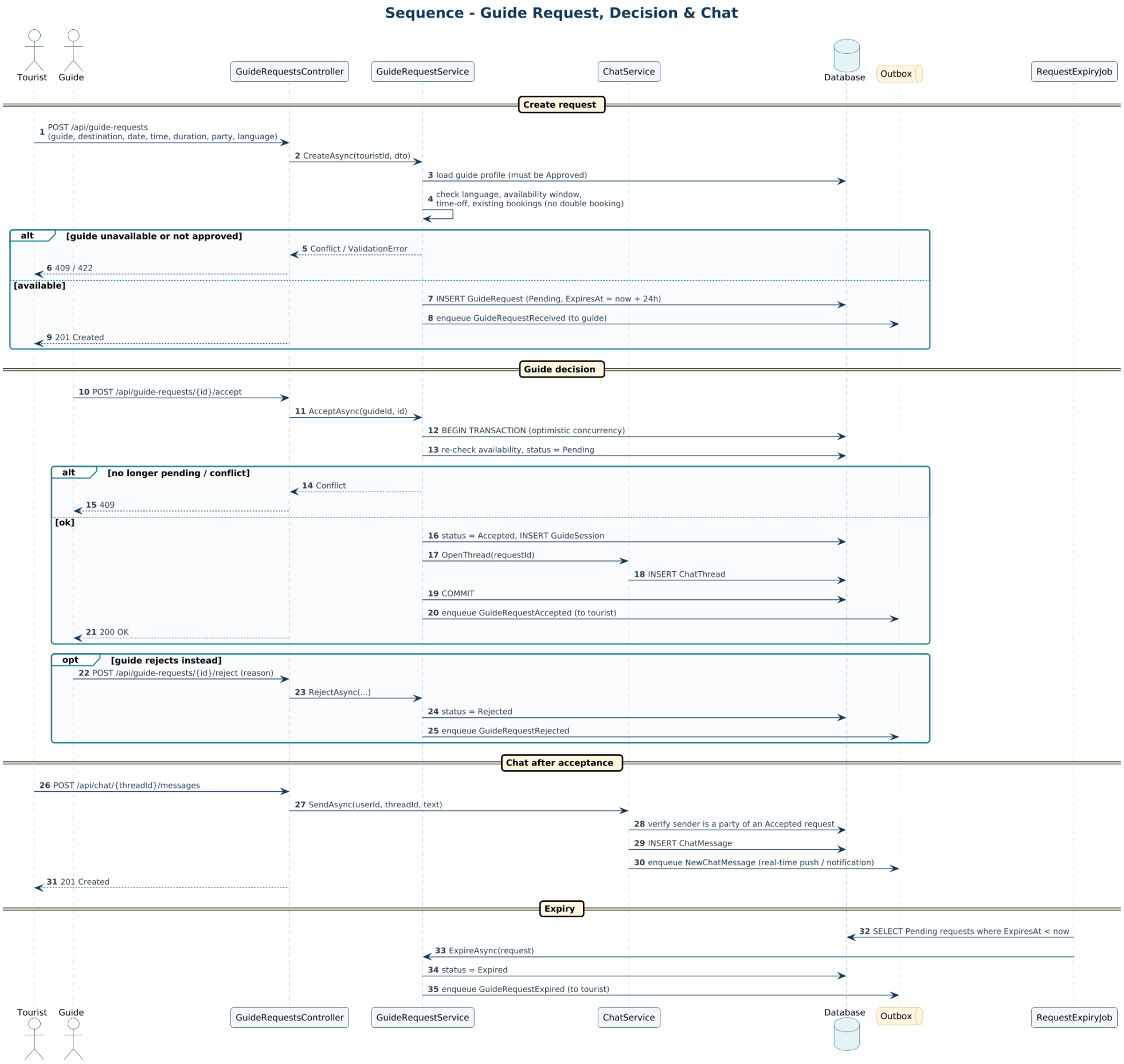


SEQ-09 Guide Request, Decision & Chat

Creating a request, guide acceptance or rejection, chat after acceptance, and automatic expiry.

Traces to: F-19, F-25

- Acceptance re-checks availability in a transaction with optimistic concurrency.
- A background job expires unanswered requests.

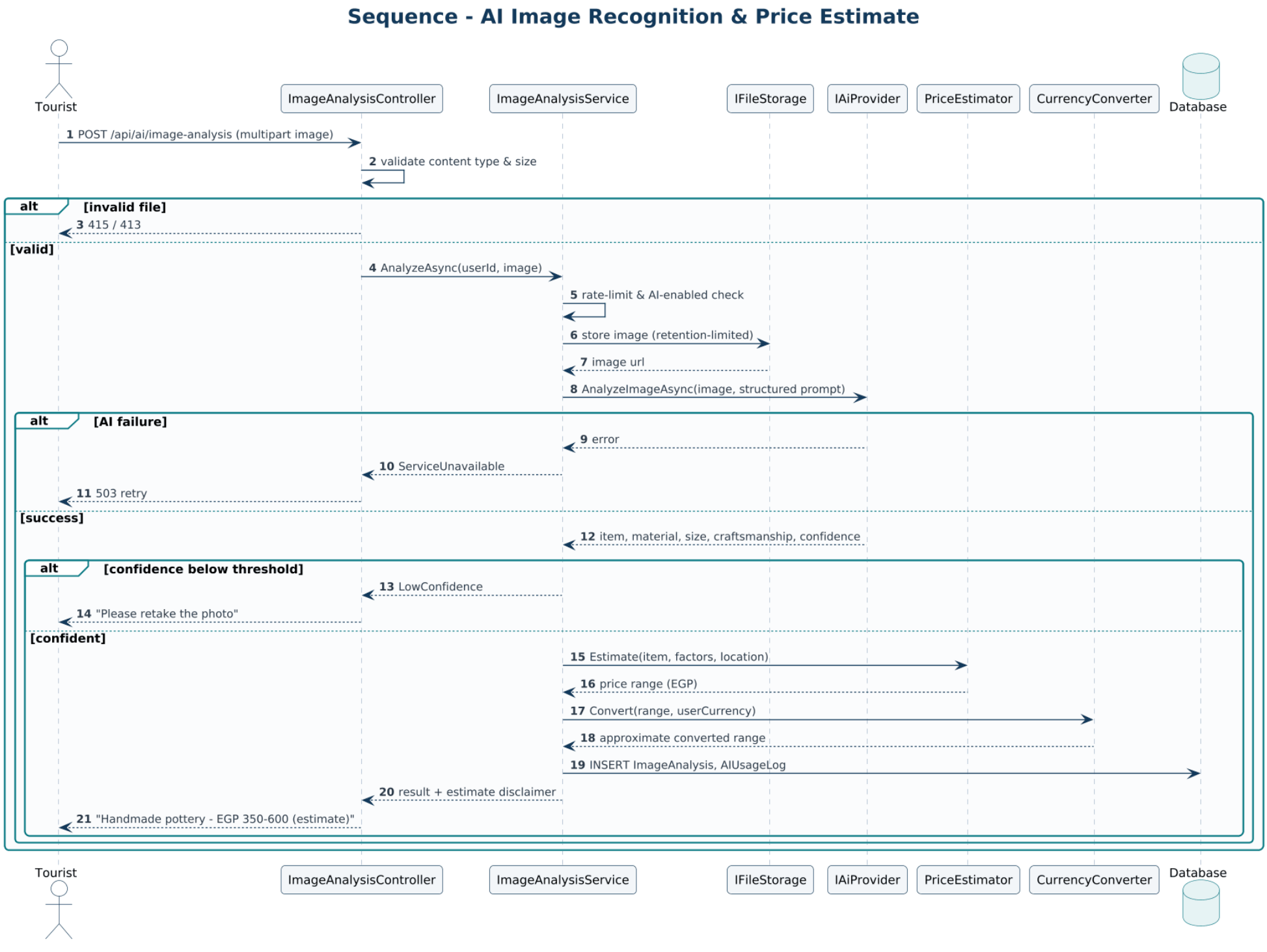


SEQ-10 AI Image Recognition & Price Estimate

Validating and analyzing an uploaded image and returning an estimated price range.

Traces to: F-17, F-27

- Low-confidence results ask for a new photo instead of guessing.
- Prices are always presented as estimates.

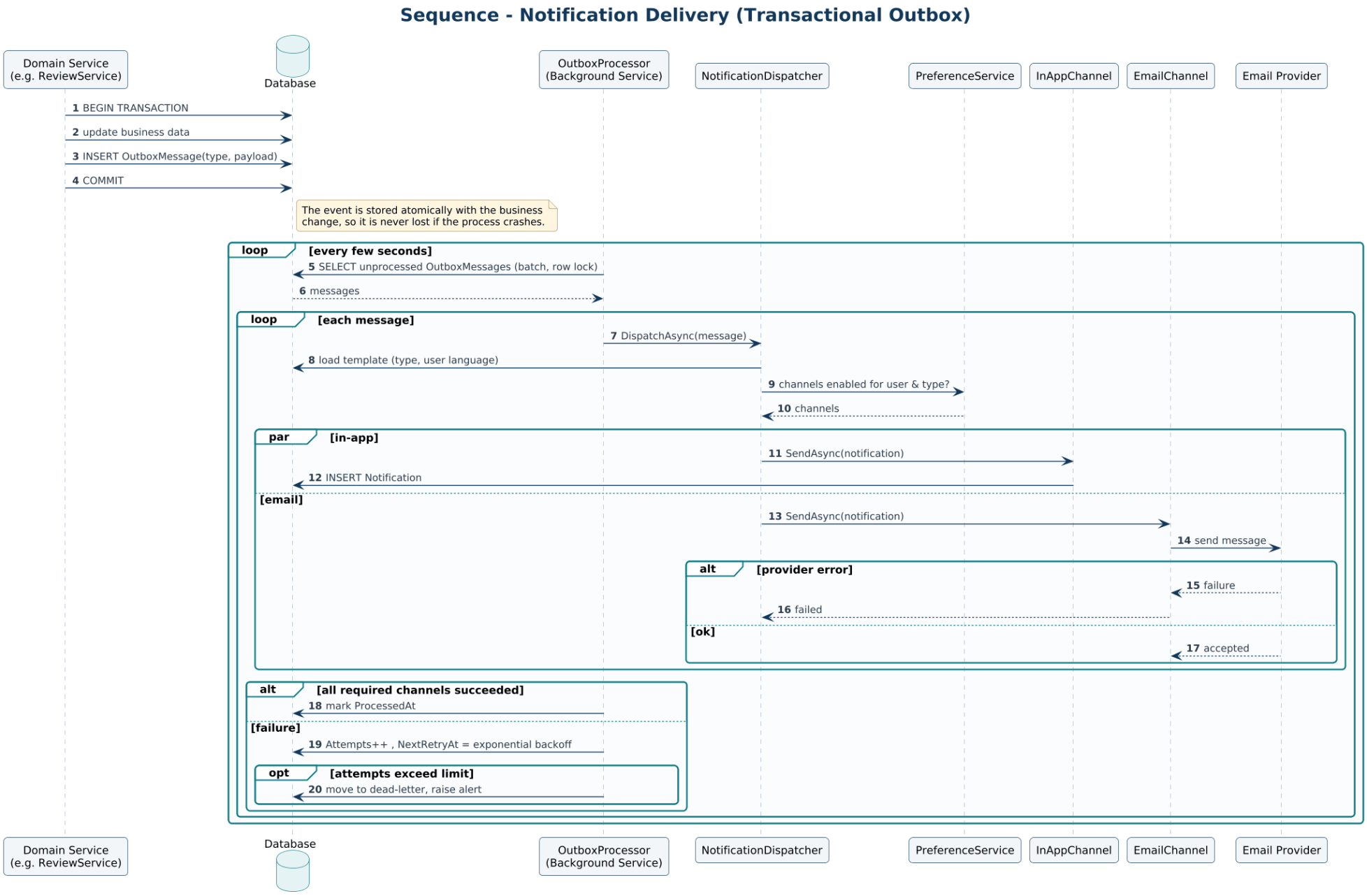


SEQ-11 Notification Delivery (Transactional Outbox)

Reliable event delivery to in-app and email channels using an outbox and a background processor.

Traces to: F-25

- The business change and the event are stored in one transaction.
- Failures retry with exponential backoff and end in a dead-letter state.

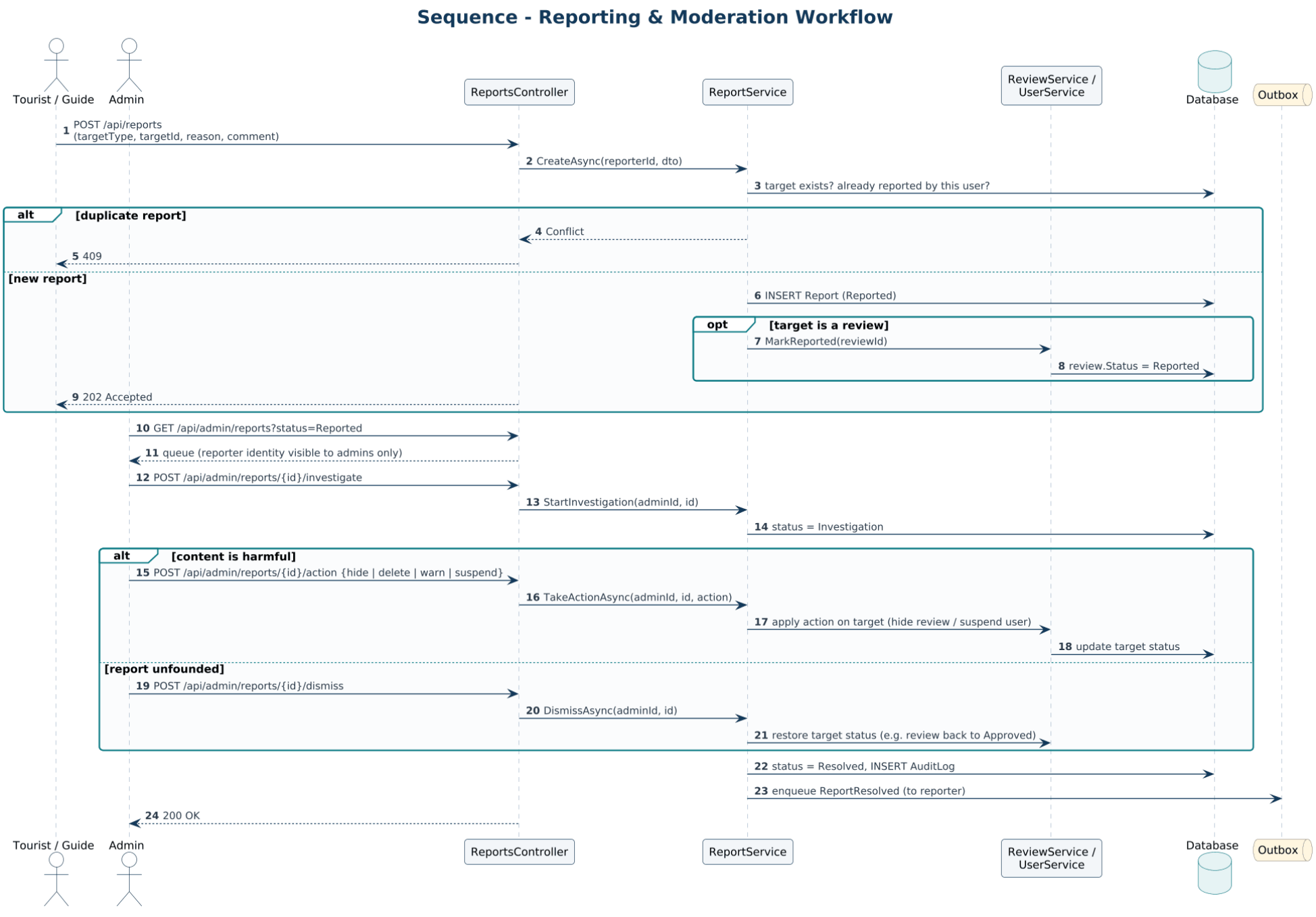


SEQ-12 Reporting & Moderation

Filing a report, investigation, admin action, and resolution.

Traces to: F-10, F-29

- Reporting a review flags it immediately; admins decide the final state.
- Every decision is written to the audit log.

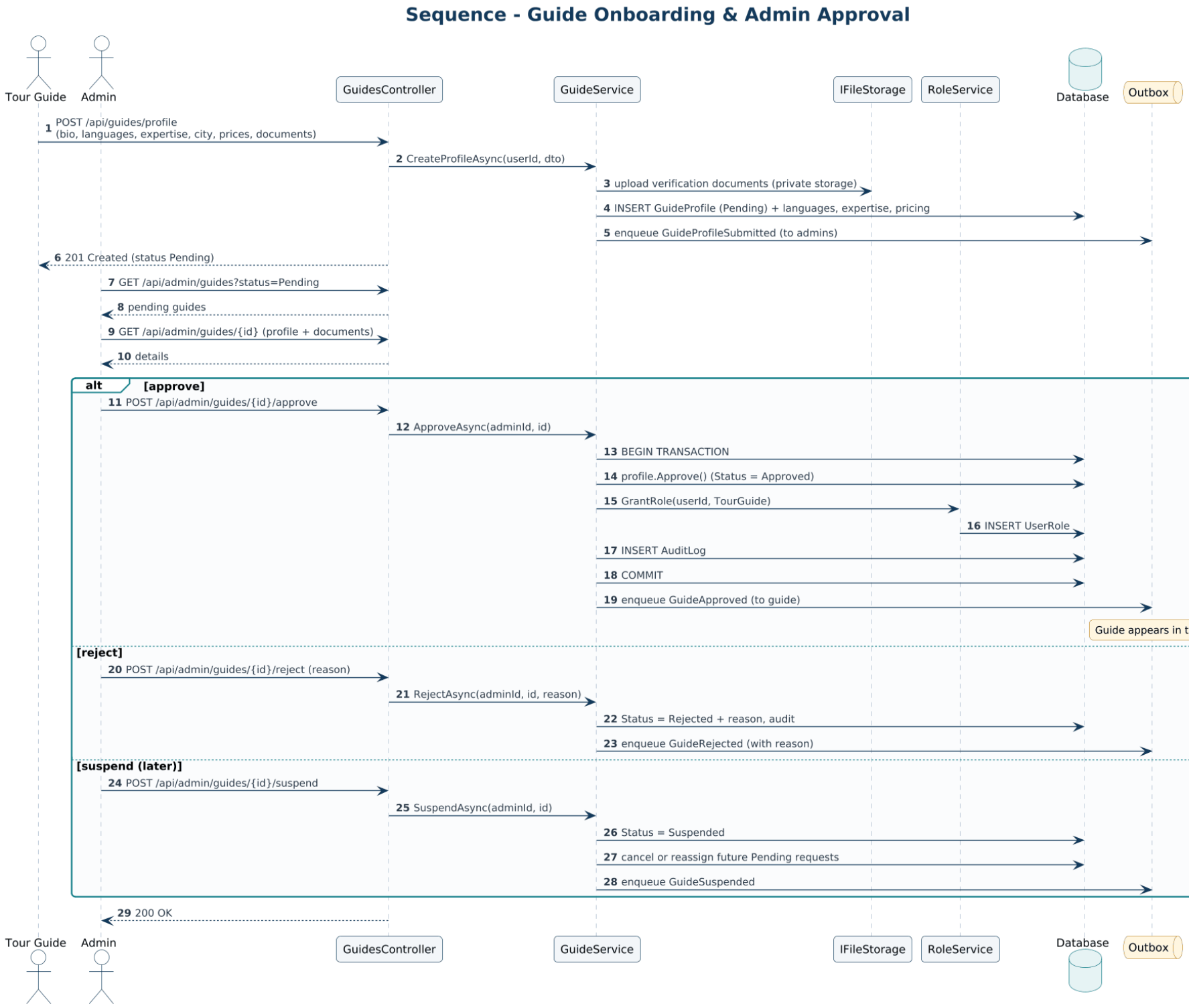


SEQ-13 Guide Onboarding & Admin Approval

Guide profile submission and the admin decision, including role assignment.

Traces to: F-01, F-18

- Approval and role grant happen atomically.
- Verification documents are kept in private storage.



Part D – Activity Diagrams

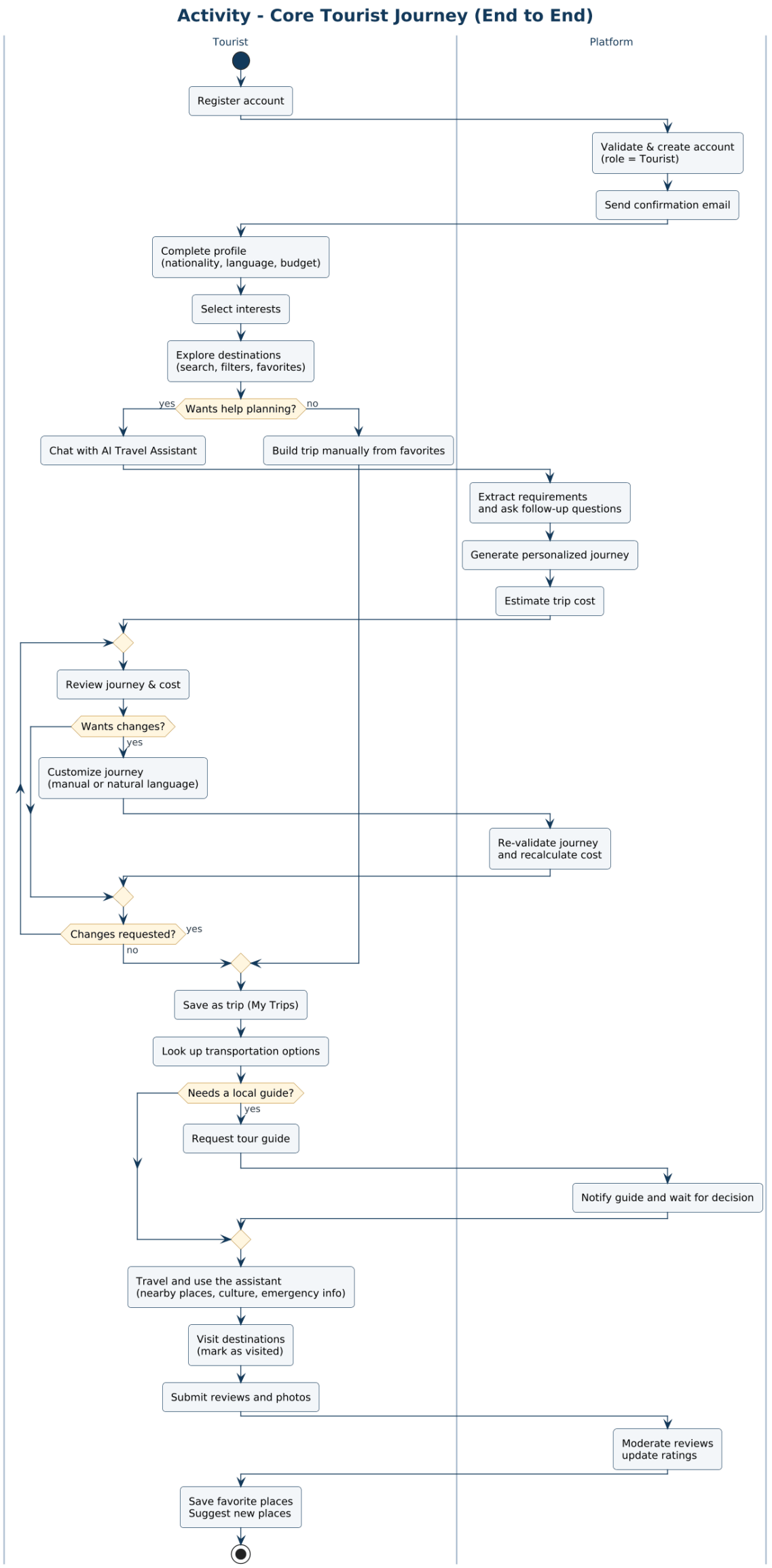
Business and algorithmic workflows, including swimlanes for responsibilities.

ACT-01 Core Tourist Journey (End to End)

The complete tourist experience from registration to reviews, with the platform steps that support it.

Traces to: F-01, F-02, F-07, F-09, F-10, F-12–F-15, F-19

- Mirrors the core user journey in the PRD.

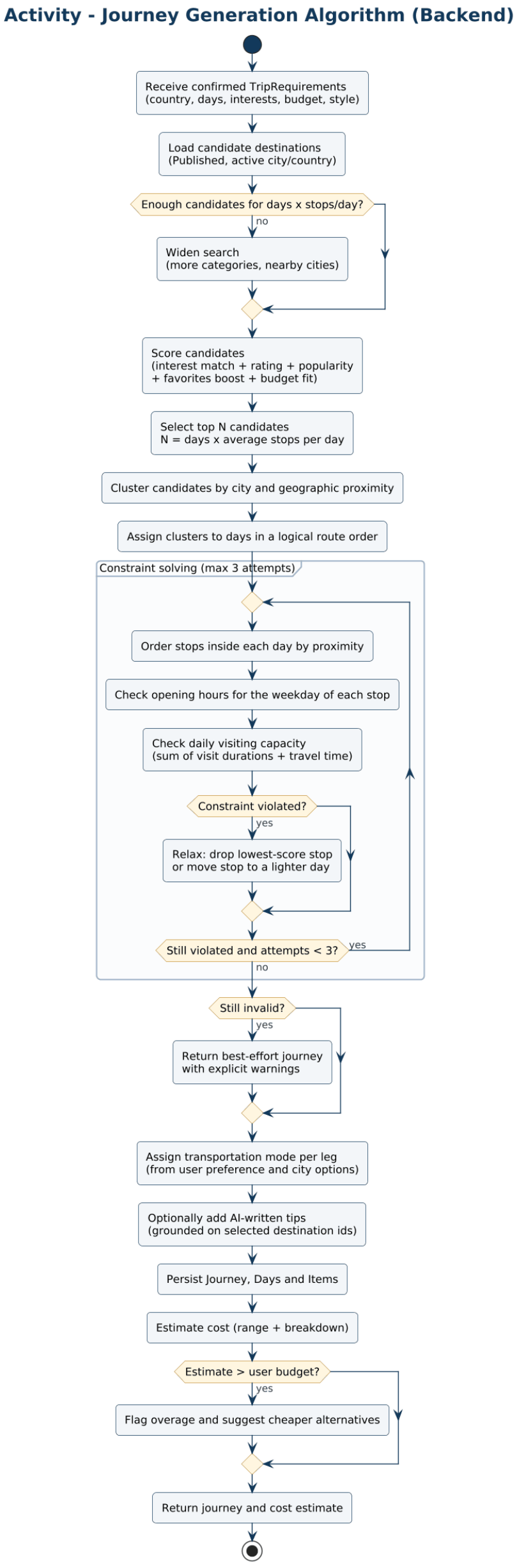


ACT-02 Journey Generation Algorithm

The backend algorithm that turns requirements into a feasible itinerary.

Traces to: F-13, F-14

- Constraint solving retries up to three times, dropping the lowest-scoring stops first.

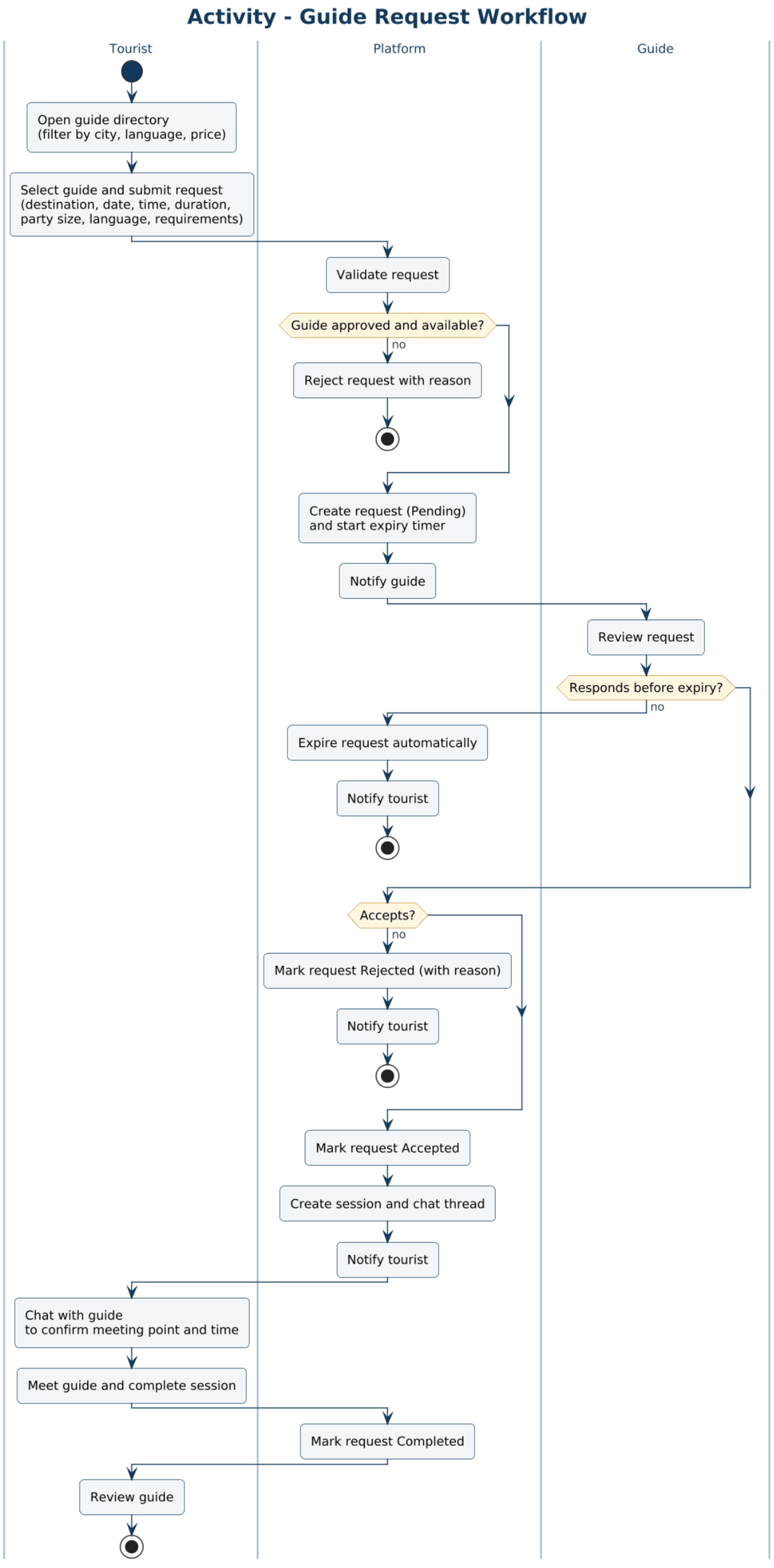


ACT-03 Guide Request Workflow

The guide request lifecycle across tourist, platform, and guide.

Traces to: F-18, F-19

- Every unhappy path ends with a notification to the tourist.

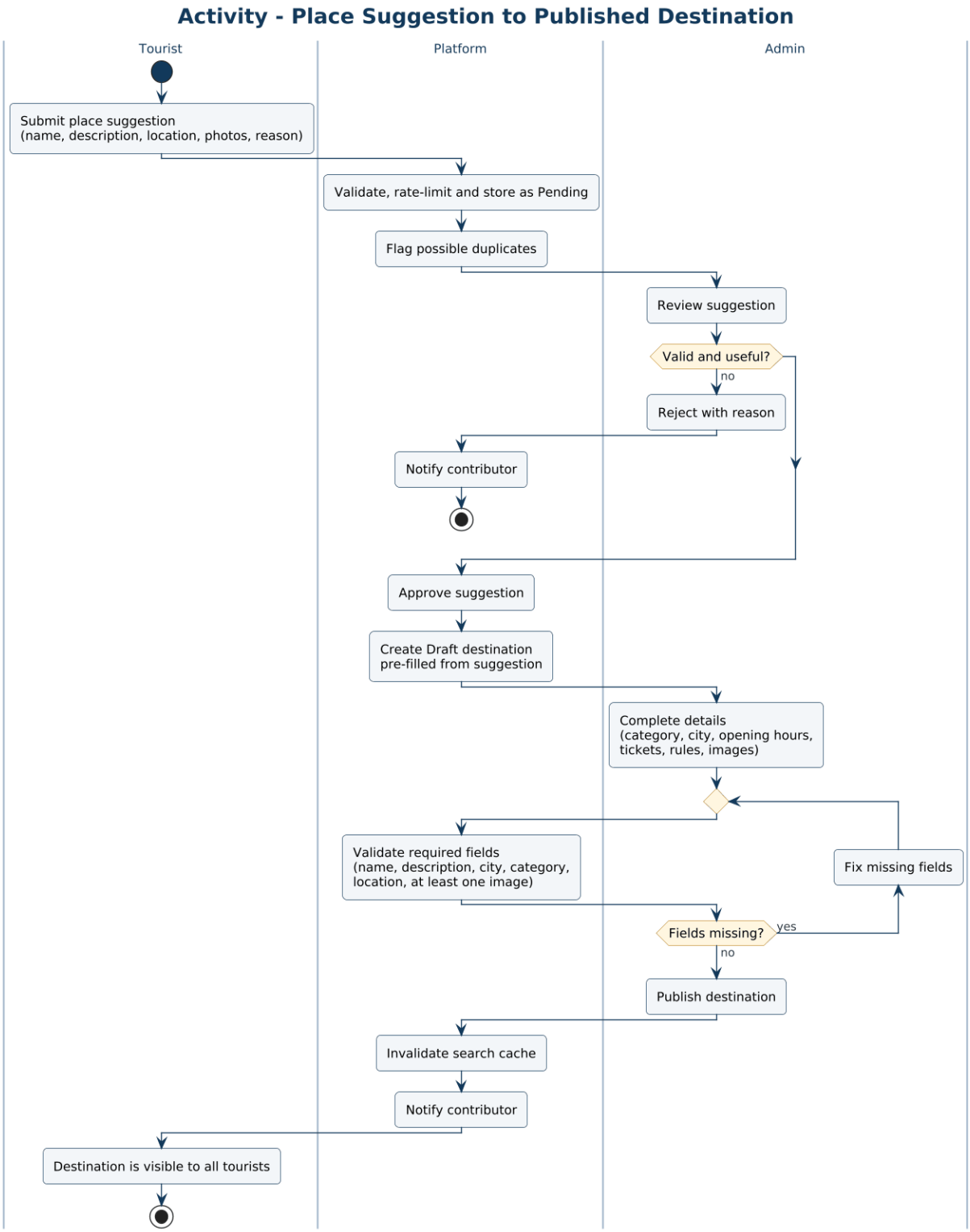


ACT-04 Place Suggestion to Published Destination

How a tourist suggestion becomes a published destination.

Traces to: F-06, F-11

- Publishing is blocked until all required fields are complete.



Part E – State Machine Diagrams

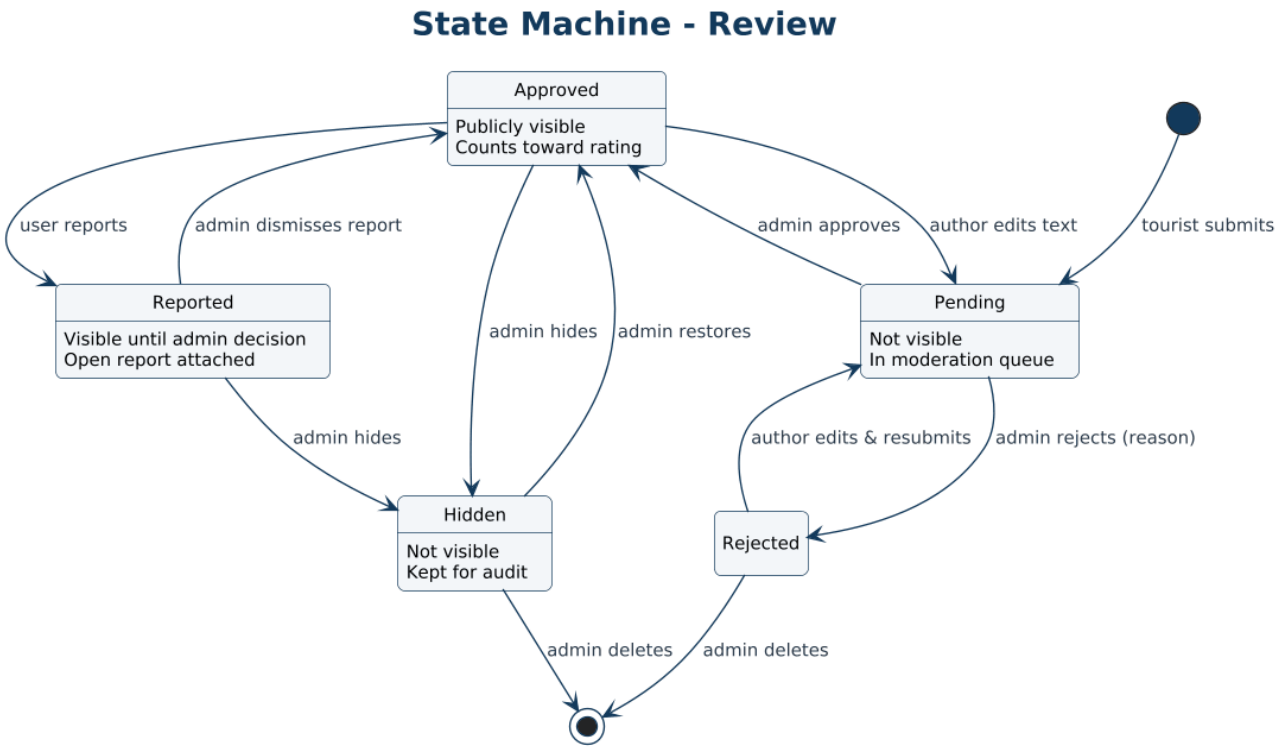
Lifecycle states and allowed transitions for the entities that carry workflow.

ST-01 Review

Moderation lifecycle of a review.

Traces to: F-10, F-29

- Only Approved reviews are public and counted in ratings.
- Editing an approved review sends it back to moderation.

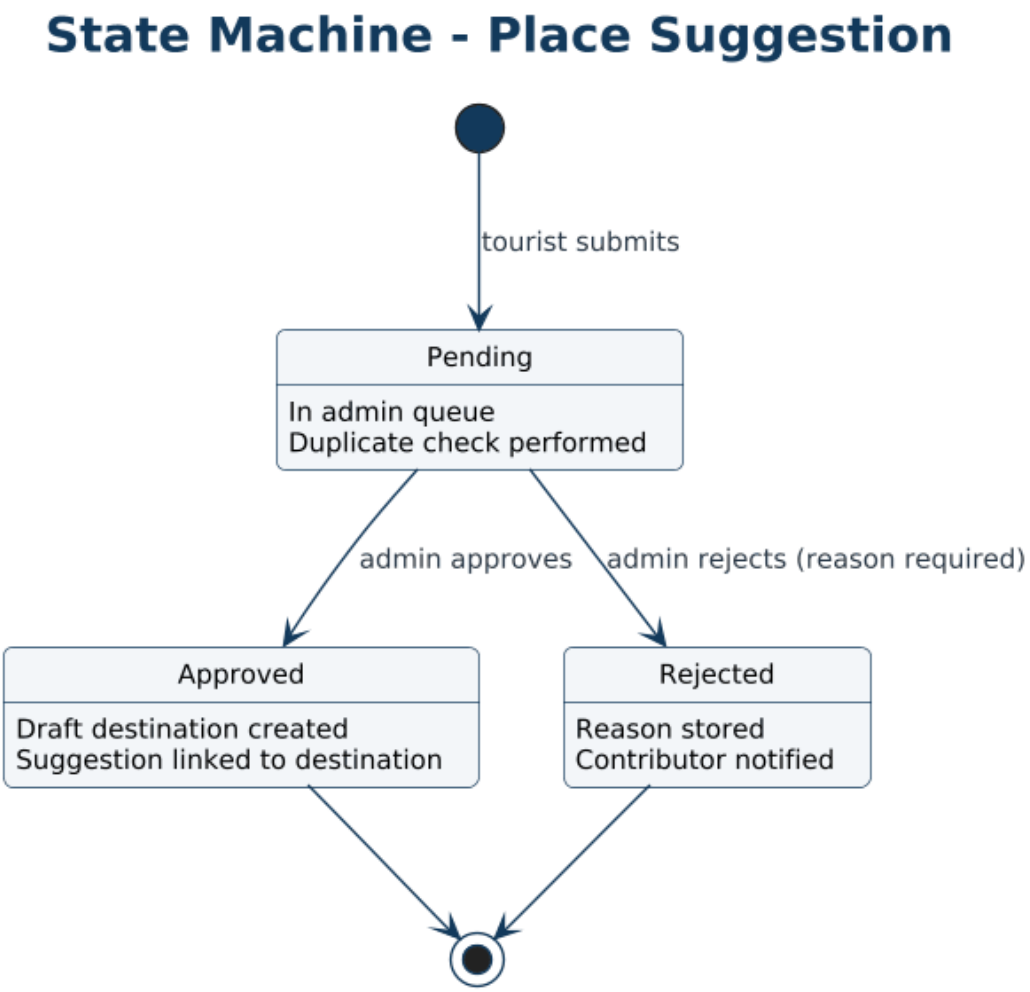


ST-02 Place Suggestion

Lifecycle of a tourist place suggestion.

Traces to: F-11

- Approval produces a Draft destination; rejection requires a reason.

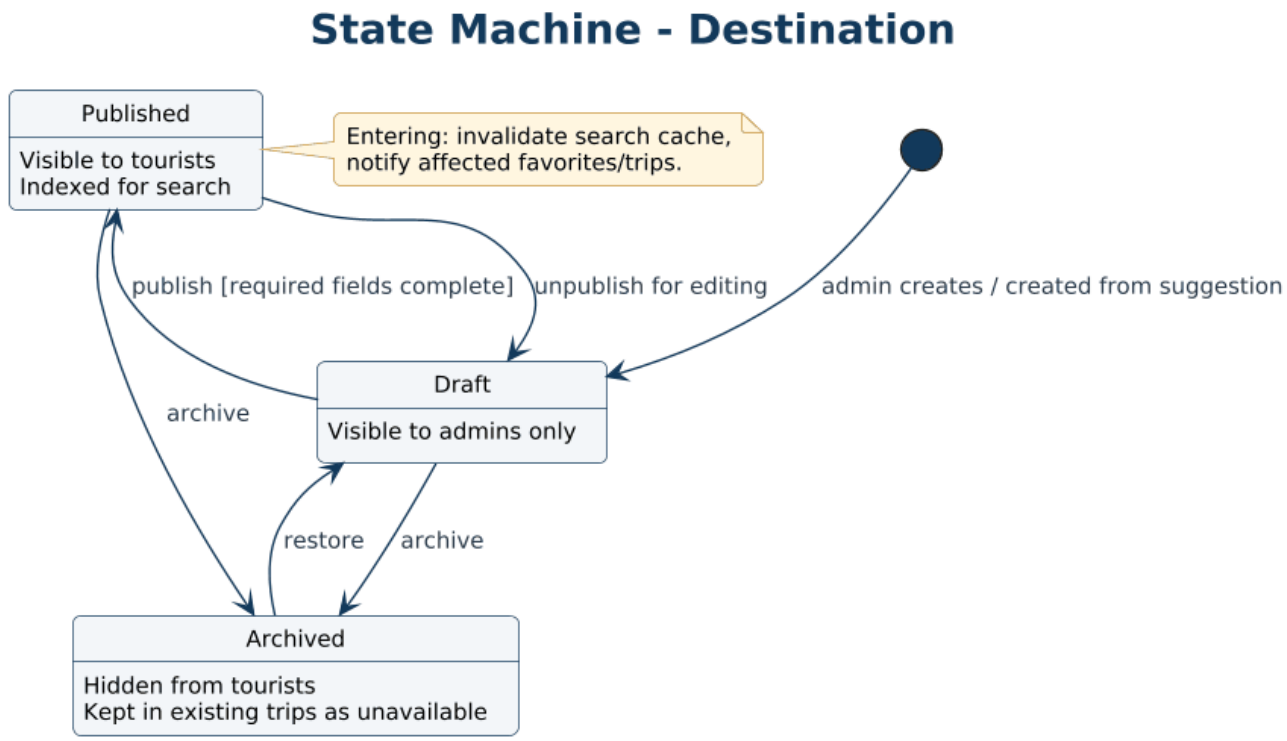


ST-03 Destination

Publication lifecycle of a destination.

Traces to: F-06

- Archived destinations stay in existing trips but are hidden from search.

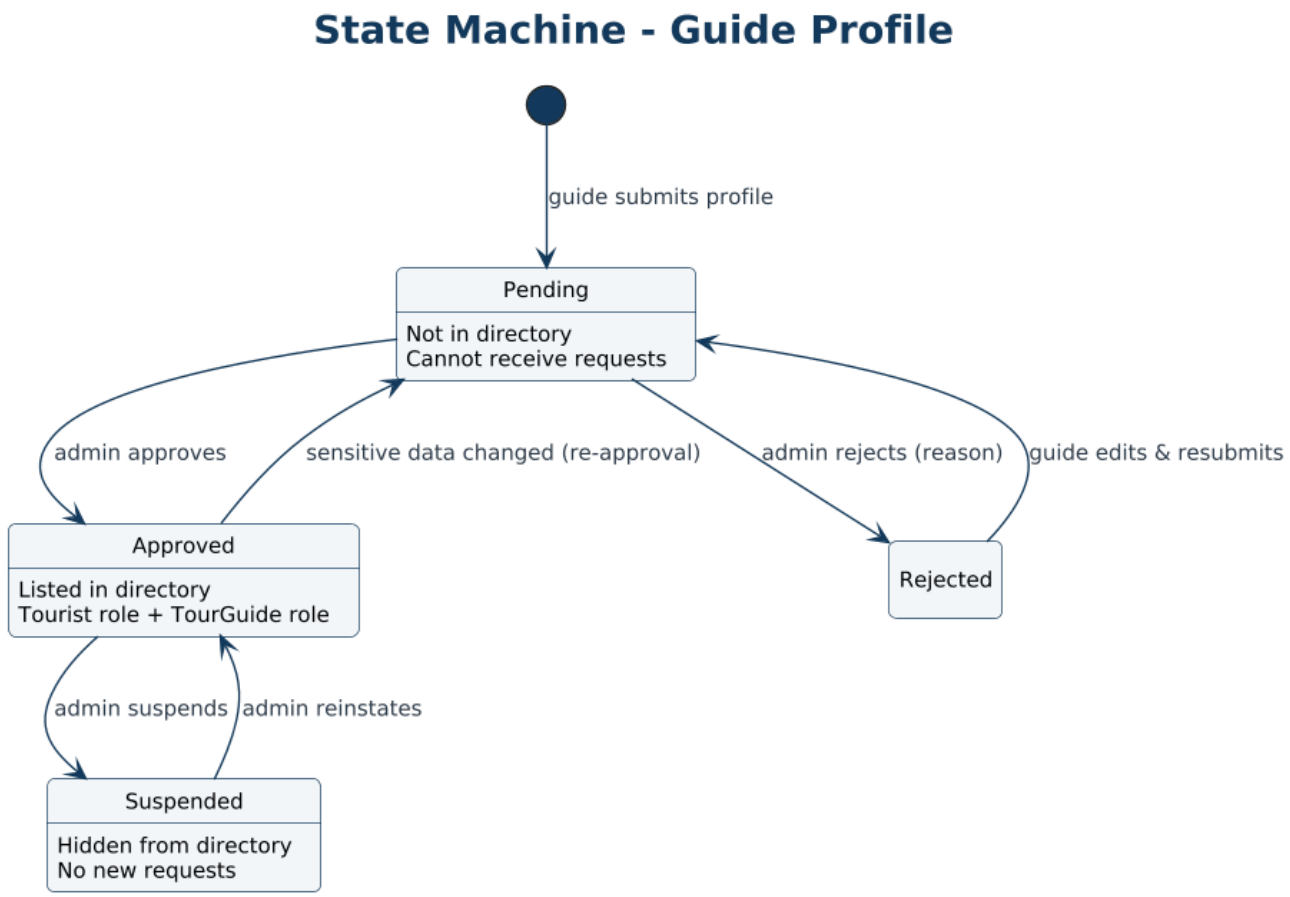


ST-04 Guide Profile

Approval lifecycle of a guide profile.

Traces to: F-18

- Changes to verification-sensitive data return the profile to Pending.

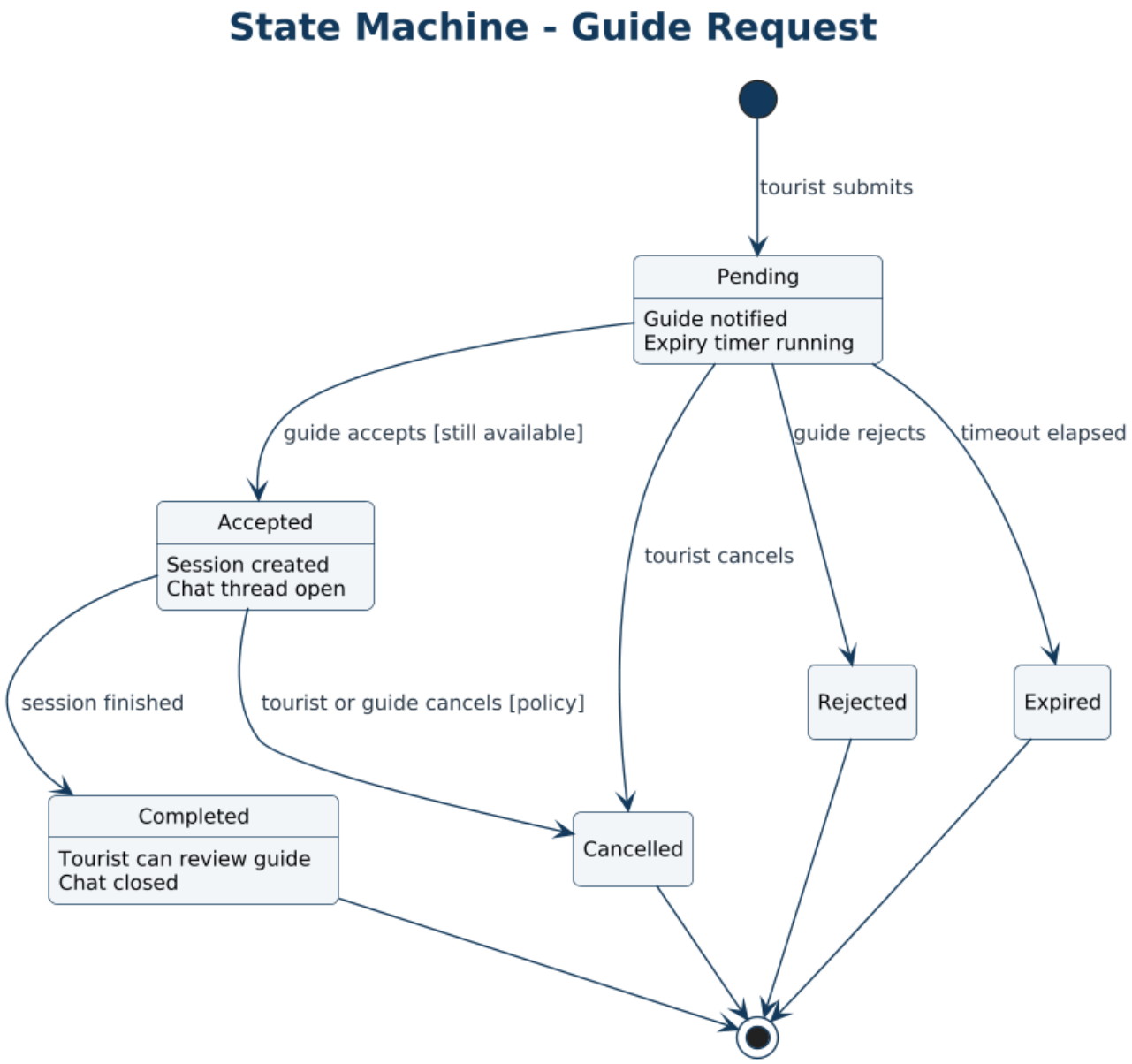


ST-05 Guide Request

Lifecycle of a guide request.

Traces to: F-19

- Pending requests expire automatically; only accepted requests can be completed.

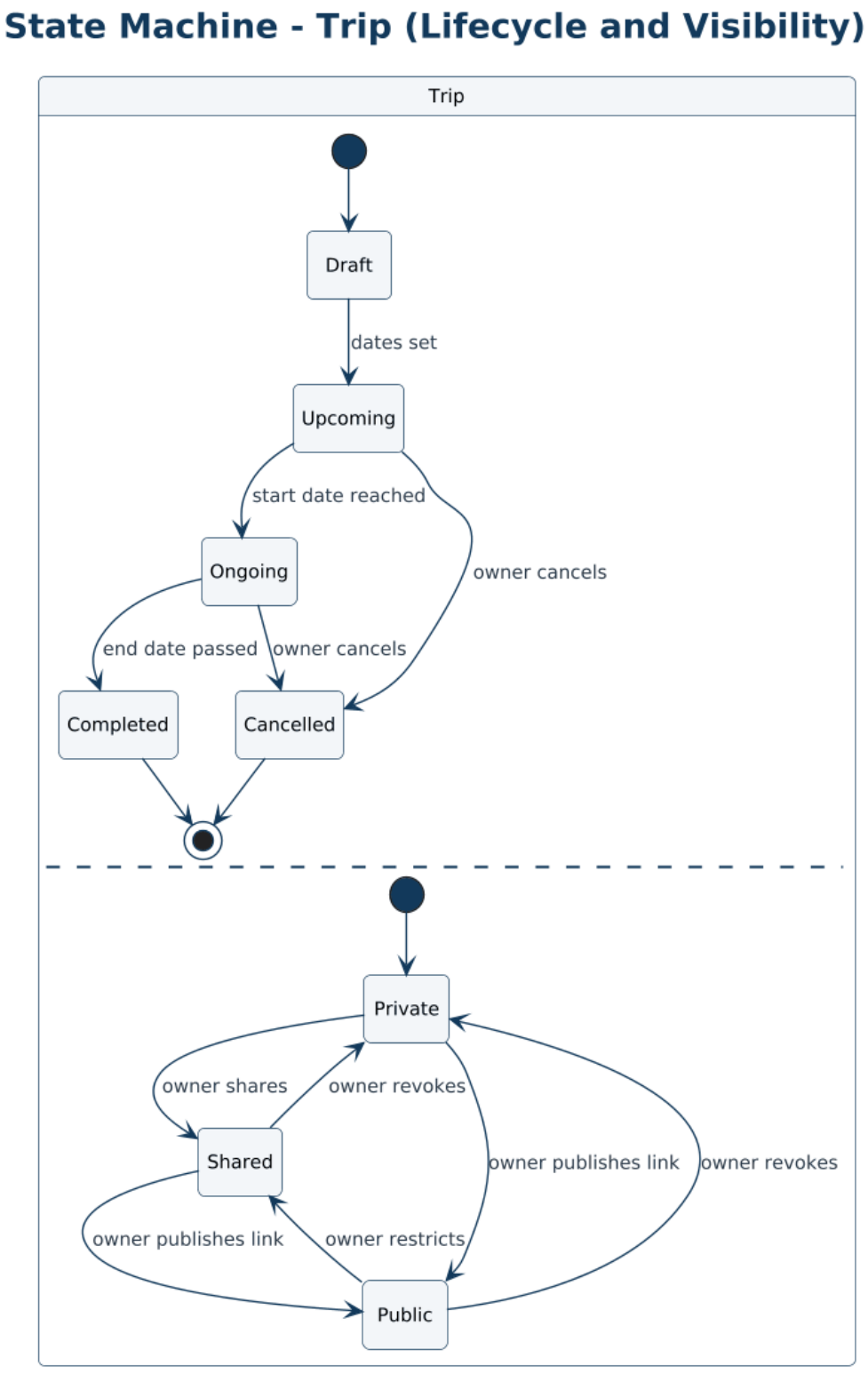


ST-06 Trip (Lifecycle and Visibility)

Trip status and its independent sharing visibility.

Traces to: F-15

- Lifecycle and visibility are orthogonal, shown as concurrent regions.

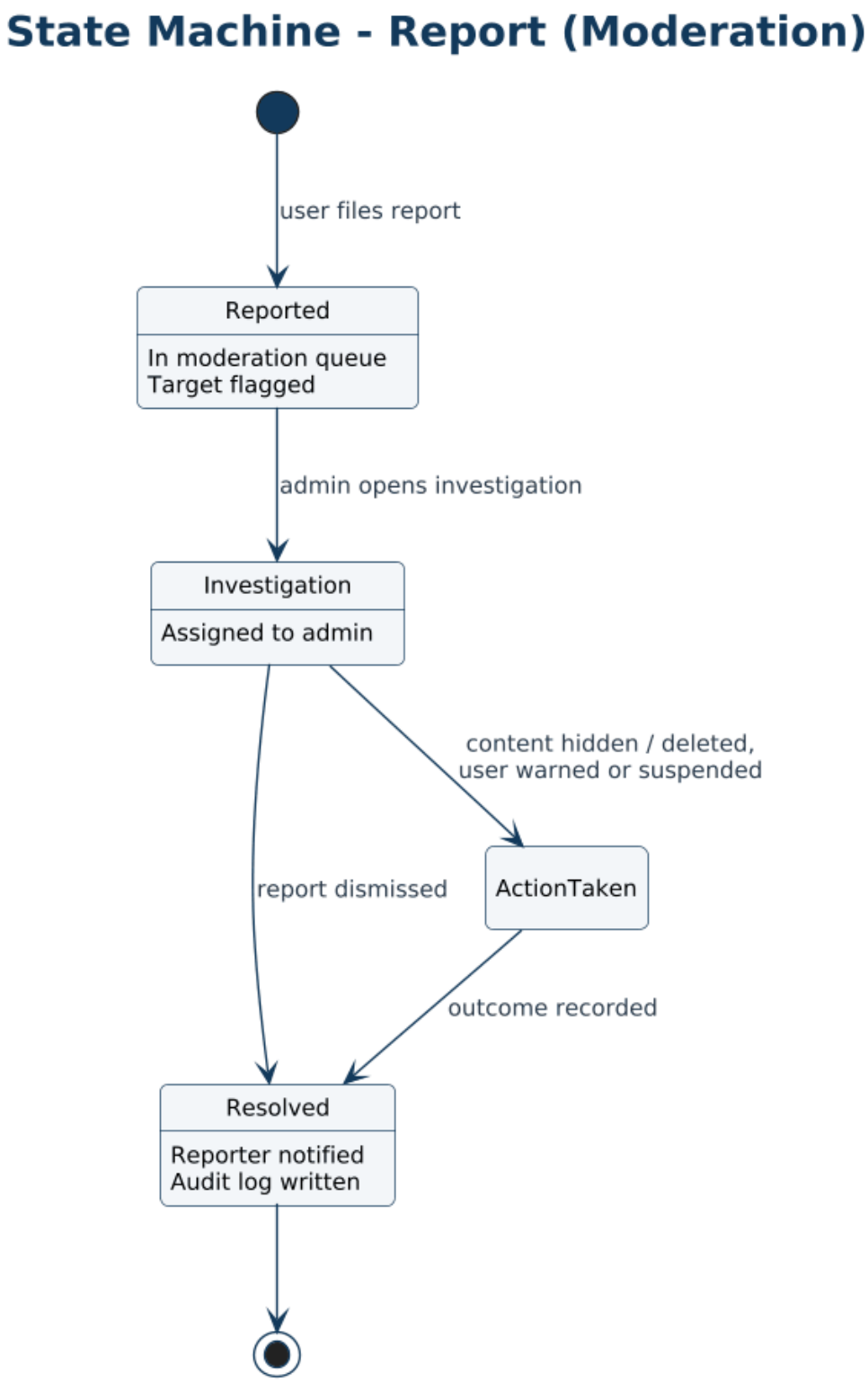


ST-07 Report

Moderation workflow of a report.

Traces to: F-29

- Every resolution is recorded in the audit log.

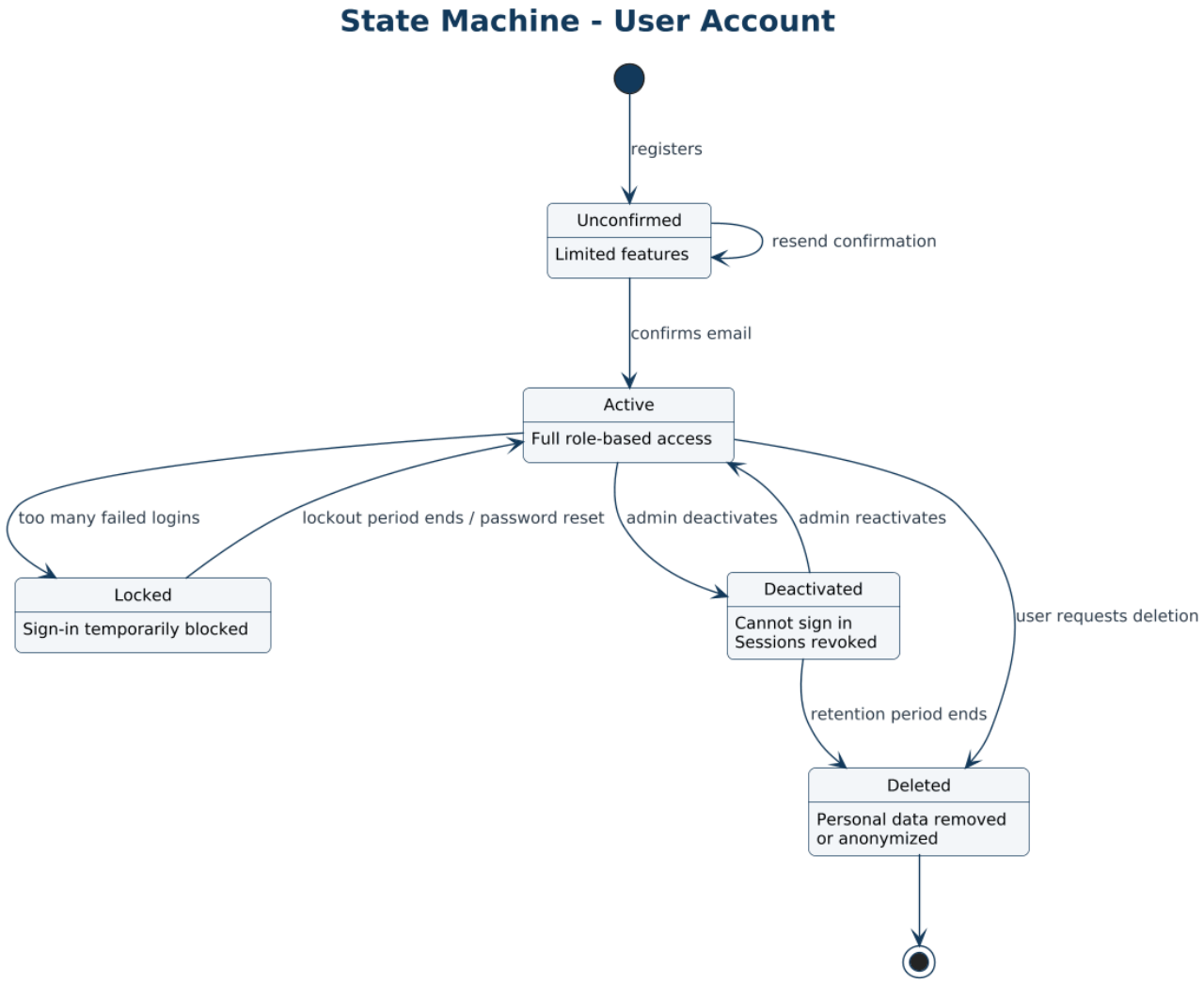


ST-08 User Account

Account states from registration to deletion.

Traces to: F-01, F-31

- Deactivation revokes all sessions immediately.



Part F – Architecture Diagrams

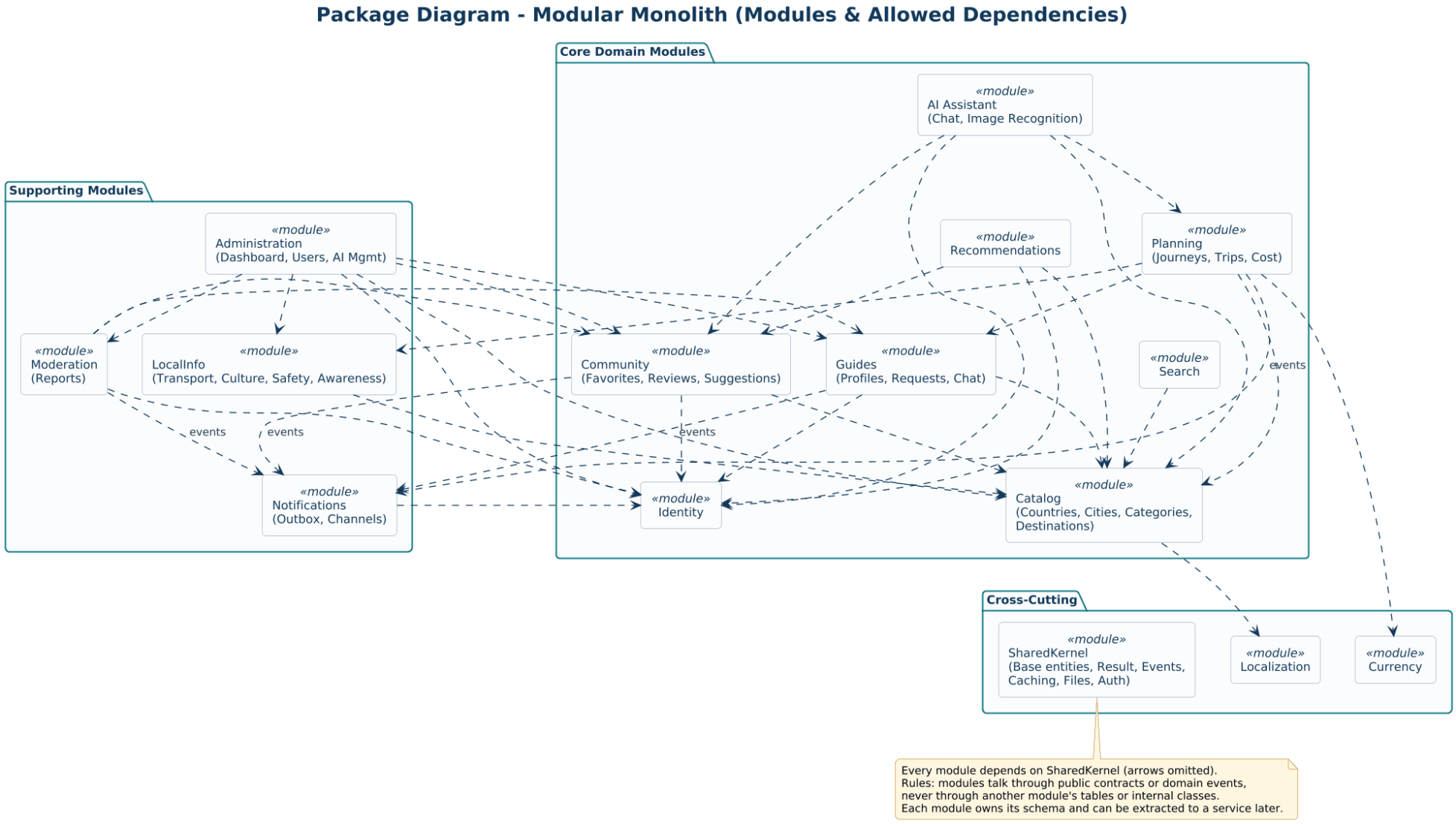
Module structure, logical components, and the proposed deployment topology.

ARCH-01 Package Diagram – Modular Monolith

Modules, their grouping, and the allowed dependency directions.

Traces to: F-01, F-03, F-06, F-10, F-13, F-18, F-25, F-30

- Modules talk through contracts or events, never through another module's tables.
- Each module can be extracted into a service later without redesign.

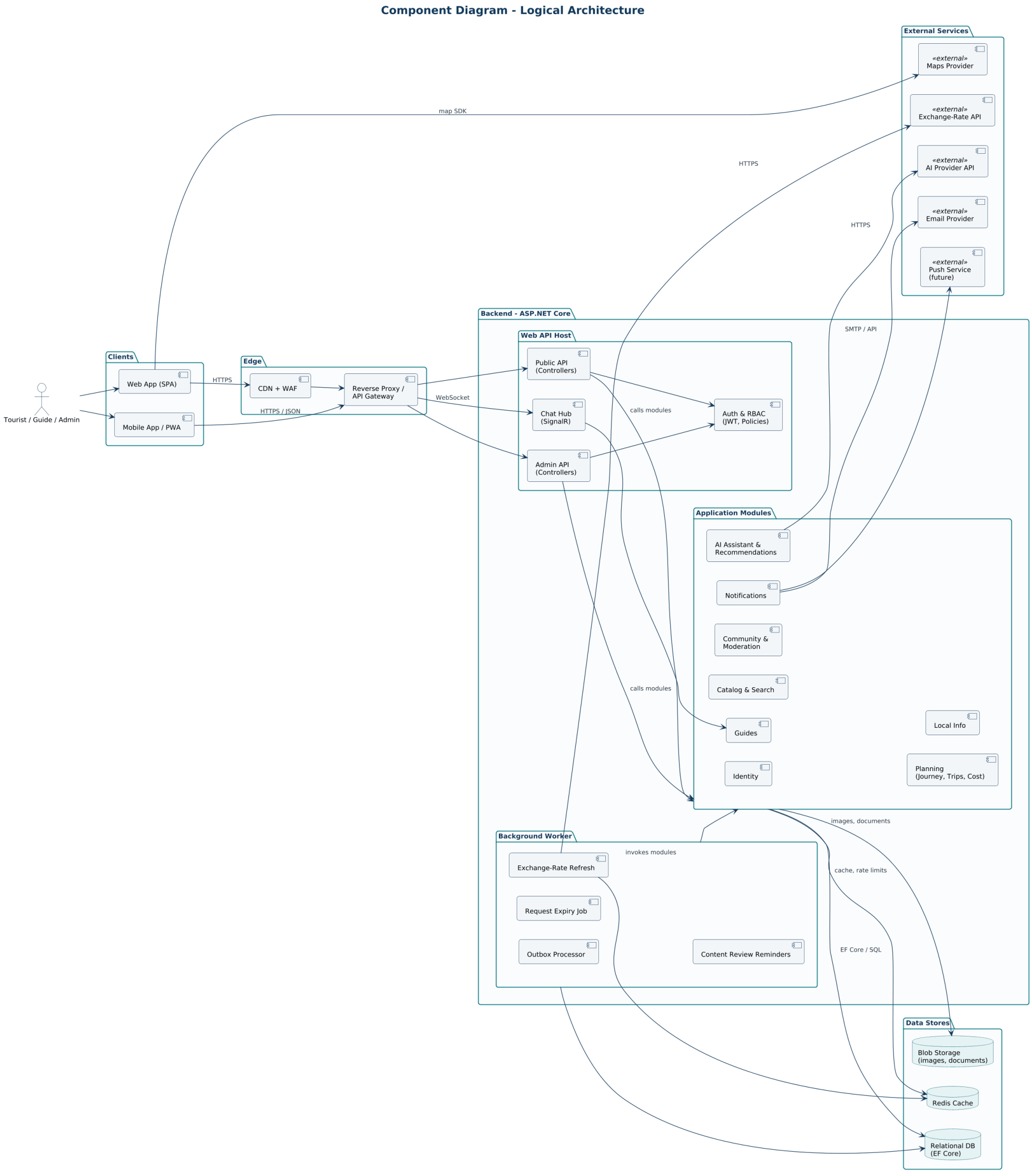


ARCH-02 Component Diagram – Logical Architecture

Clients, edge, API host, application modules, background worker, data stores, and external services.

Traces to: F-01, F-12, F-19, F-25, F-27, F-30

- Public and Admin APIs share modules but use separate authorization policies.
- The worker handles outbox delivery, request expiry, and exchange-rate refresh.

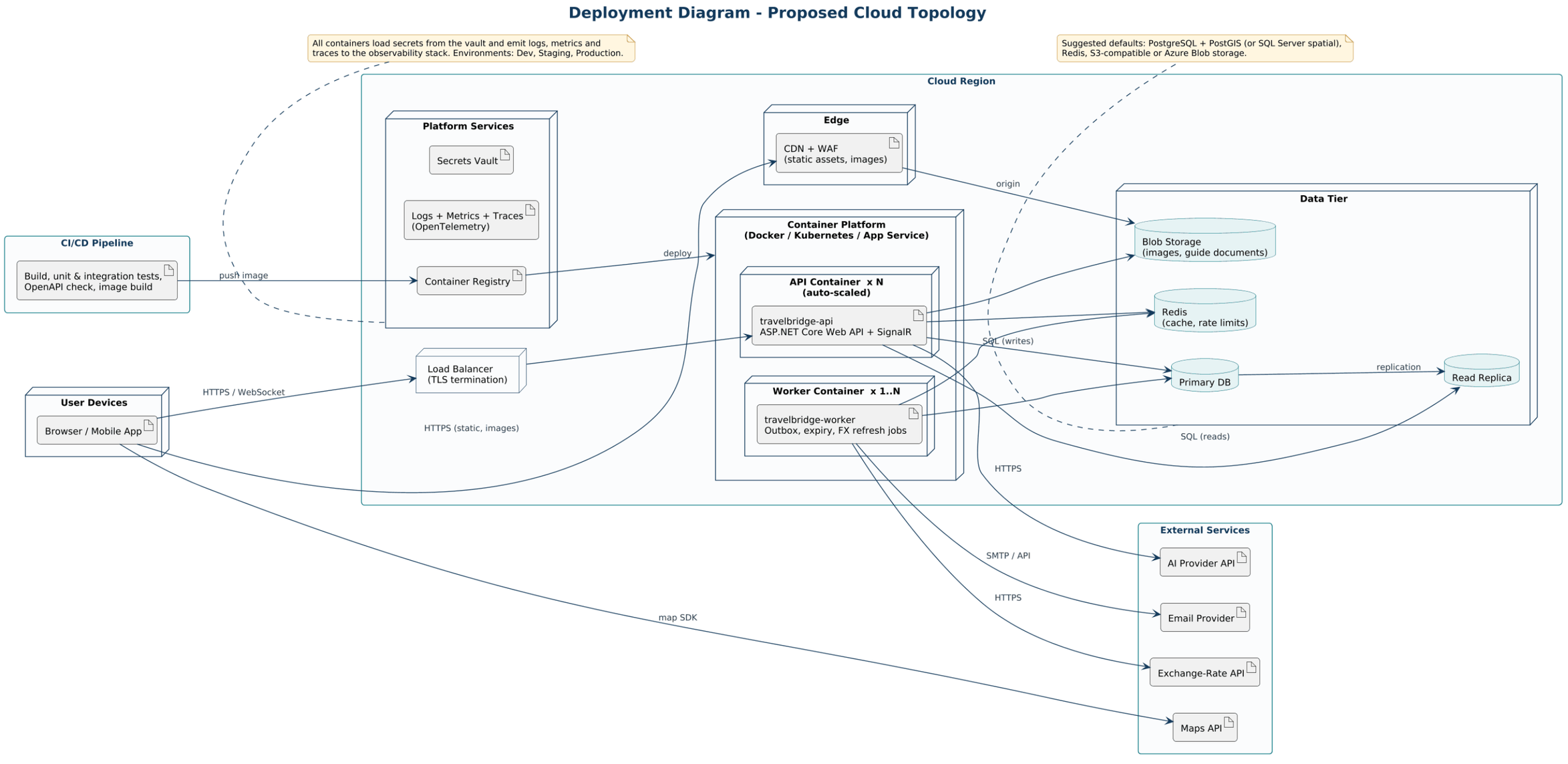


ARCH-03 Deployment Diagram – Proposed Topology

How the system is deployed: edge, load balancer, containers, data tier, platform services, CI/CD, and external providers.

Traces to: F-30

- API containers are stateless and scale horizontally.
- Read queries can use a replica; secrets come from a vault.



Part G – Entity-Relationship Diagrams

Physical data model. Types follow PostgreSQL naming and can be mapped to SQL Server equivalents.

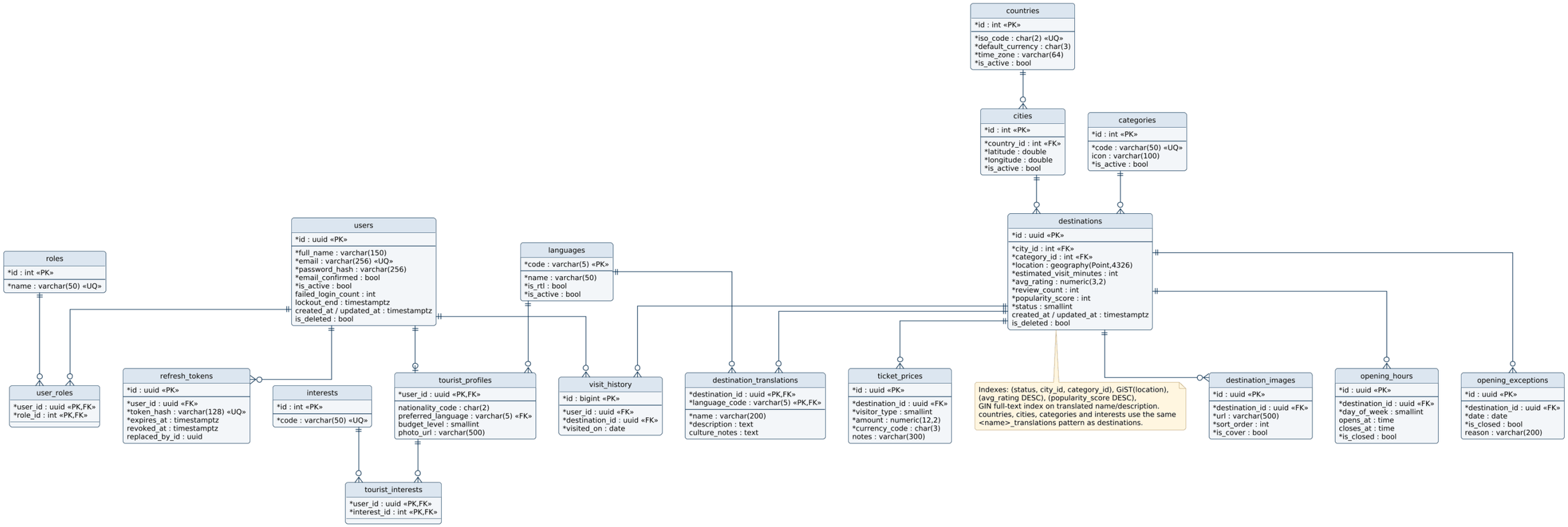
ERD-01 Identity, Geography & Catalog

Tables, keys, and relationships for accounts, geography, and destination content.

Traces to: F-01–F-06, F-26

- Public entities use UUID keys; small reference tables use integer keys.
- Translation tables give every language its own row with English fallback.

ERD (1/4) - Identity, Geography & Catalog



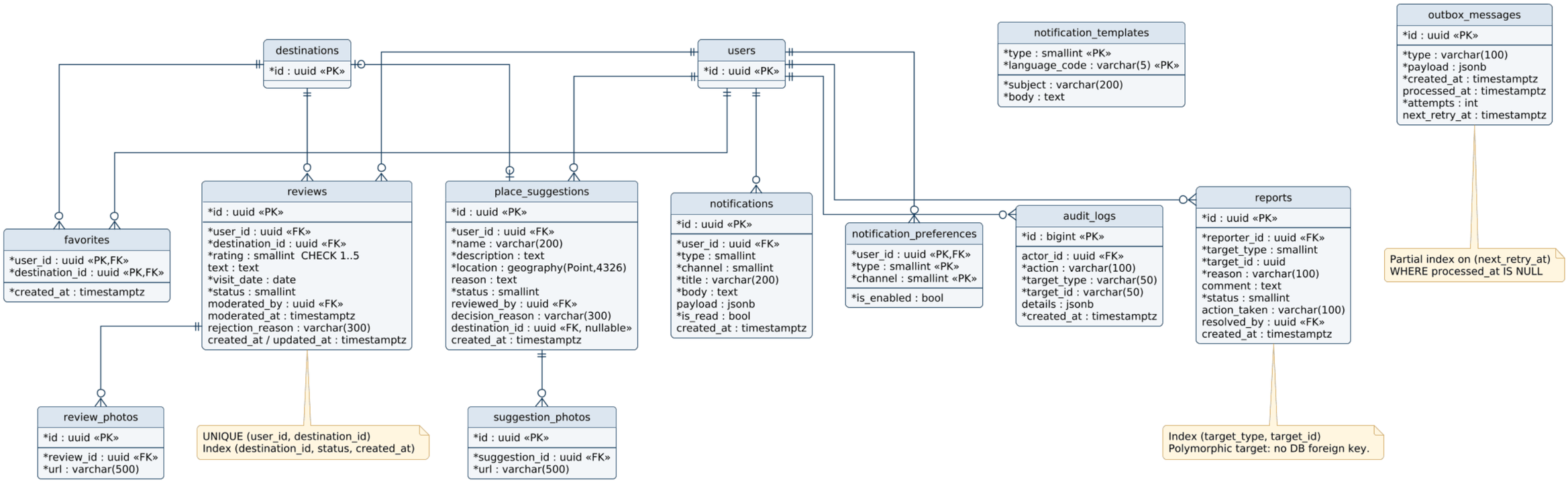
ERD-02 Community, Moderation, Notifications & Audit

Tables for favorites, reviews, suggestions, reports, notifications, the outbox, and audit logs.

Traces to: F-09–F-11, F-25, F-29

- One review per user per destination is enforced by a unique constraint.
- Reports use a polymorphic target, so integrity is enforced in the application layer.

ERD (2/4) - Community, Moderation, Notifications & Audit

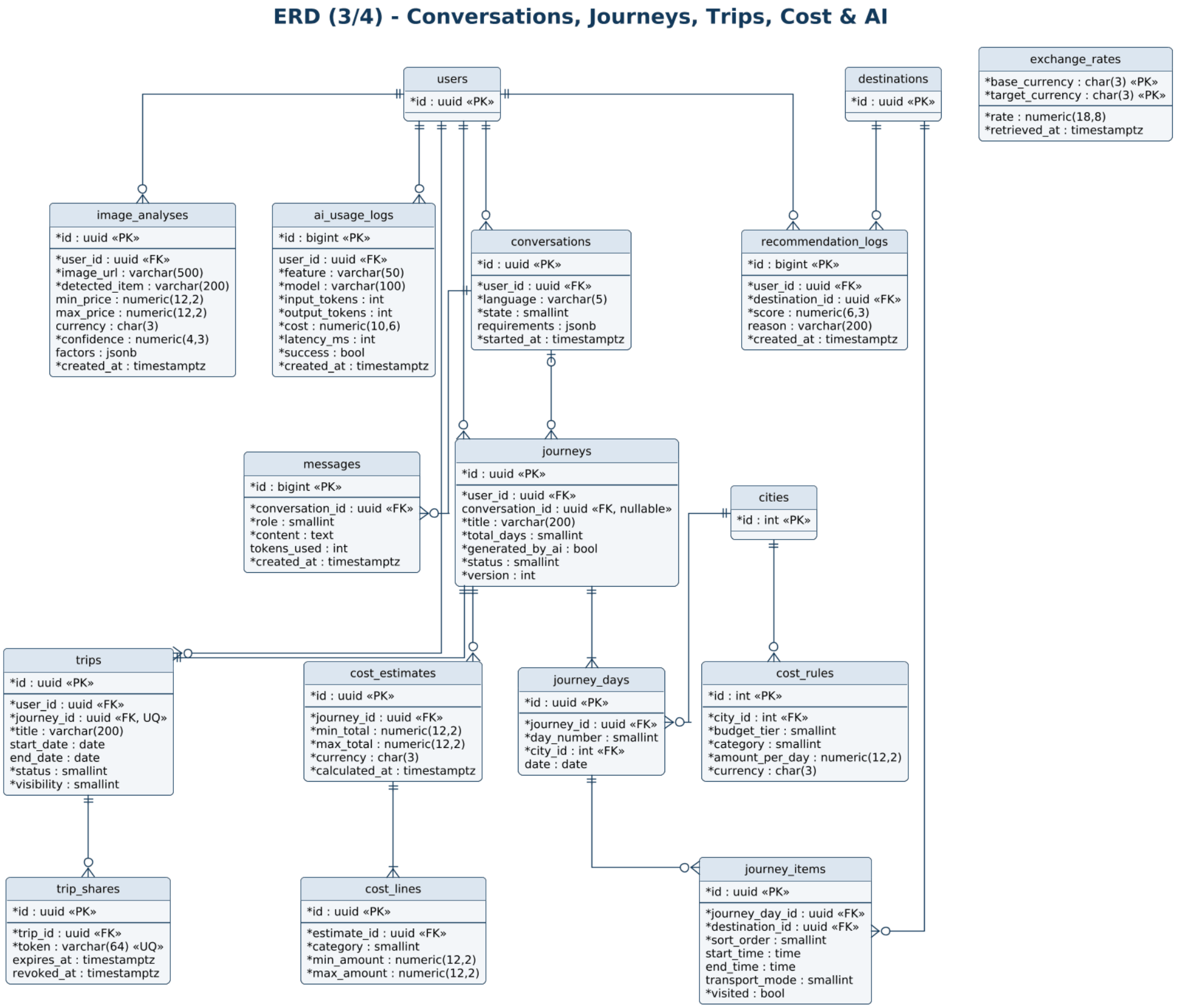


ERD-03 Conversations, Journeys, Trips, Cost & AI

Tables for AI conversations, itineraries, trips, cost estimates, exchange rates, and AI usage logs.

Traces to: F-12–F-17, F-27

- Cost estimates are stored as snapshots to keep history stable.
- Journeys carry a version number to support safe concurrent edits.

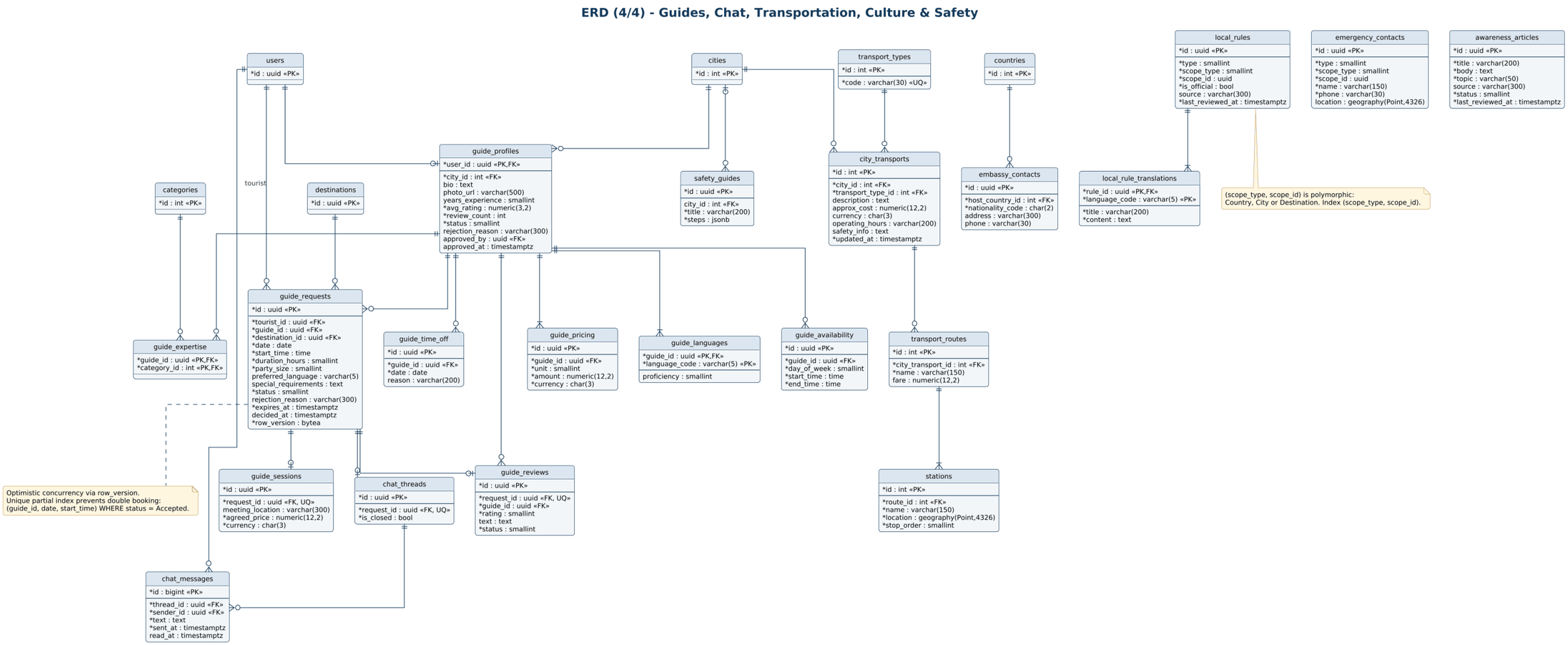


ERD-04 Guides, Chat, Transportation, Culture & Safety

Tables for guides, requests, chat, transportation, rules, emergency data, and awareness content.

Traces to: F-18–F-20, F-22–F-24

- A partial unique index prevents double-booking an accepted guide slot.
- Rules and emergency contacts use a (scope_type, scope_id) pair.



Appendix A. Feature-to-Diagram Traceability

Which diagrams describe each feature from the requirements specification.

Feature	Name	Diagrams
F-01	Authentication & Authorization	UC-01, CLS-01, SEQ-01, SEQ-13, ACT-01, ST-08, ARCH-01, ARCH-02, ERD-01
F-02	Tourist Profile & Preferences	UC-02, CLS-01, ACT-01, ERD-01
F-03	Countries Management	UC-01, UC-05, CLS-01, ARCH-01, ERD-01
F-04	Cities	UC-01, UC-05, CLS-01, ERD-01
F-05	Destination Categories	UC-05, CLS-01, ERD-01
F-06	Destinations	UC-01, UC-05, CLS-01, CLS-06, SEQ-04, ACT-04, ST-03, ARCH-01, ERD-01
F-07	Explore & Search	UC-01, CLS-06, SEQ-02, ACT-01
F-08	Destination Details	UC-01, CLS-06
F-09	Favorites / Wishlist	UC-02, CLS-02, ACT-01, ERD-02
F-10	Reviews & Feedback	UC-02, UC-06, CLS-02, SEQ-03, SEQ-12, ACT-01, ST-01, ARCH-01, ERD-02
F-11	Place Suggestions	UC-02, UC-06, CLS-02, SEQ-04, ACT-04, ST-02, ERD-02
F-12	AI Travel Assistant	UC-03, CLS-03, CLS-07, SEQ-05, ACT-01, ARCH-02, ERD-03
F-13	Journey Generation & Customization	UC-03, CLS-03, CLS-07, SEQ-06, SEQ-07, ACT-01, ACT-02, ARCH-01, ERD-03
F-14	Trip Cost Estimation	UC-03, CLS-03, SEQ-06, SEQ-07, SEQ-08, ACT-01, ACT-02, ERD-03
F-15	My Trips & Trip Sharing	UC-03, CLS-03, ACT-01, ST-06, ERD-03
F-16	Recommendation Engine	UC-03, CLS-03, CLS-07, ERD-03
F-17	AI Image Recognition	UC-03, CLS-03, CLS-07, SEQ-10, ERD-03
F-18	Tour Guide Profiles & Approval	UC-02, UC-04, UC-06, CLS-04, SEQ-13, ACT-03, ST-04, ARCH-01, ERD-04
F-19	Guide Requests & Communication	UC-02, UC-04, CLS-04, SEQ-09, ACT-01, ACT-03, ST-05, ARCH-02, ERD-04
F-20	Transportation Information	UC-01, UC-05, CLS-05, ERD-04
F-21	Interactive Map	UC-01
F-22	Culture, Customs & Local Rules	UC-01, UC-05, CLS-05, ERD-04
F-23	Safety & Emergency Information	UC-01, UC-05, CLS-05, ERD-04
F-24	Tourist Awareness	UC-01, UC-05, CLS-05, ERD-04
F-25	Notifications	UC-04, CLS-02, CLS-08, SEQ-03, SEQ-09, SEQ-11, ARCH-01, ARCH-02, ERD-02
F-26	Multi-Language Support	CLS-01, ERD-01
F-27	Currency Support	UC-03, CLS-08, SEQ-08, SEQ-10, ARCH-02, ERD-03
F-28	Offline Travel Information	UC-03
F-29	Reporting & Moderation	UC-02, UC-04, UC-06, CLS-02, SEQ-12, ST-01, ST-07, ERD-02
F-30	Admin Dashboard & Metrics	UC-06, ARCH-01, ARCH-02, ARCH-03
F-31	Admin User Management	UC-06, ST-08

Feature	Name	Diagrams
F-32	Admin AI Management	UC-06

Appendix B. Regenerating the Diagrams

All diagrams are written in PlantUML so they can be versioned and reviewed like code. A shared style file (style.puml) is included by every diagram.

- Install Java 17 or later and Graphviz.
- Download plantuml.jar from the PlantUML releases page.
- Run: java -DPLANTUML_LIMIT_SIZE=8192 -jar plantuml.jar -tpng *.puml (use -tsvg for vector output).
- Edit the .puml file, re-render, and commit both source and output.

Most editors (VS Code, JetBrains IDEs) also have PlantUML plugins for live preview.